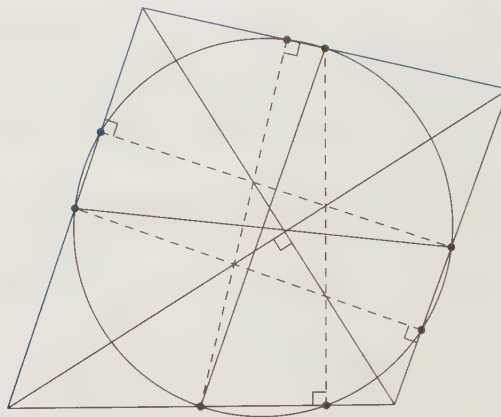




The Eight Point Circle Theorem

It is easier to square the circle than to get round a mathematician. – Augustus De Morgan

8

CHAPTER

Quadrilaterals

8.1 Quadrilateral Basics

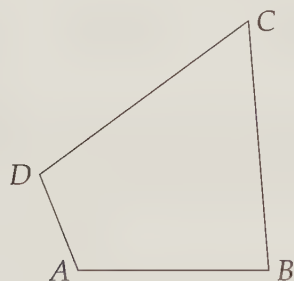


Figure 8.1: A Quadrilateral

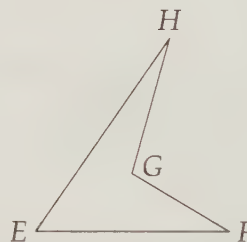


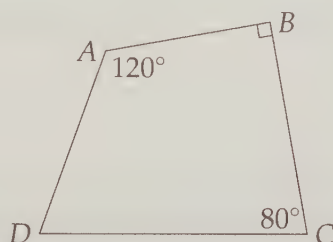
Figure 8.2: A Concave Quadrilateral

Figure 8.1 shows a **quadrilateral**, namely $ABCD$. A quadrilateral has four sides, four vertices, and four angles. All the quadrilaterals we will study in this text have all four vertices in the same plane. Nearly all quadrilaterals we deal with in this book are **convex**, like $ABCD$ above, meaning that all of the interior angles of the quadrilateral are less than 180° . There are also **concave quadrilaterals**, like $EFGH$ in Figure 8.2.

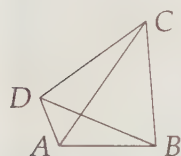
In this section we address basic definitions and facts that hold for all quadrilaterals. In later sections, we turn to specific types of quadrilaterals.

Problems

Problem 8.1: Use what you know about the sum of the angles in a triangle to find $\angle D$ in the figure below.



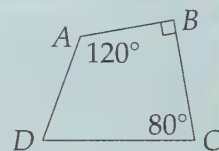
As you've probably already considered, we can connect two more pairs of vertices in a quadrilateral by connecting the opposite vertices.



At left, we have added $\overline{AC}$ and $\overline{BD}$ to quadrilateral $ABCD$. We call these segments connecting the opposite vertices of a quadrilateral the **diagonals** of a quadrilateral.

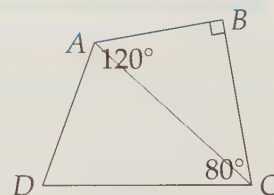
Let's see how we can use a diagonal in a problem.

Problem 8.1: Find $\angle D$ in the figure.



Solution for Problem 8.1: We don't yet know anything about angles of a quadrilateral, but we know about triangles, so we add diagonal $\overline{AC}$, thus making two triangles, $\triangle ABC$ and $\triangle ADC$. The sum of the angles in each of these triangles is 180° , so the sum of the angles in the whole quadrilateral $ABCD$ is $2(180^\circ) = 360^\circ$. Now we can quickly find $\angle D$:

$$\angle D = 360^\circ - \angle DAB - \angle ABC - \angle BCD = 70^\circ.$$



□

Important: The interior angles of any quadrilateral add to 360° .



Concept: Triangles, triangles, triangles! As you'll see in the coming sections, nearly everything we can derive for specific types of quadrilaterals stems from triangles.



Exercises

8.1.1 In quadrilateral $ABCD$, $\angle A = 100^\circ$, $\angle B = 50^\circ$, and $\angle C$ is 30° more than $\angle D$. Find $\angle D$.

8.1.2 The angles of a quadrilateral are x , $3x - 10^\circ$, 27° , and $4x - 30^\circ$. Find the measure of the largest angle of the quadrilateral.

8.1.3

- (a) If two of the angles of a quadrilateral are equal, must the other two be equal?
- (b) Is it possible for three of the angles of a quadrilateral to equal each other, but the fourth angle be different?
- (c) If all of the angles of a quadrilateral have the same measure, what is this measure?

8.1.4 How many different quadrilaterals can be formed by connecting the four points shown at right?

8.1.5 We showed earlier that the interior angles of a convex quadrilateral add to 360° . Show that the interior angles of a concave quadrilateral add to 360° as well. **Hints:** 391

8.2 Trapezoids

A **trapezoid** is a quadrilateral in which two sides are parallel. Figure 8.3 shows trapezoid $WXYZ$, in which $\overline{WX} \parallel \overline{ZY}$.

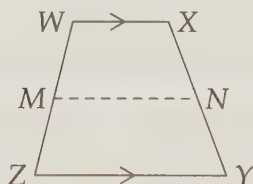


Figure 8.3: A Trapezoid

These parallel sides are often called the **bases** of the trapezoid, with the other sides called the **legs**. The segment that connects the midpoints of the legs is the **median** of the trapezoid; $\overline{MN}$ is the median of $WXYZ$ above.

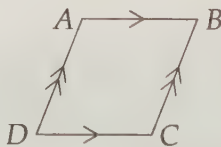


Figure 8.4: Is This a Trapezoid?

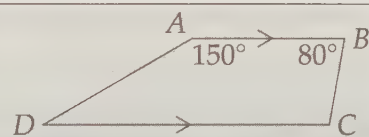
You might be wondering ‘What if there are two pairs of opposite sides that are parallel like in Figure 8.4; is the quadrilateral still a trapezoid?’ Good question! Unfortunately, there isn’t a good answer. Some people define a trapezoid as having exactly one pair of opposite sides, so Figure 8.4

would not be a trapezoid. Other people define a trapezoid as having *at least one* pair of opposite sides; to these people, Figure 8.4 would be considered a trapezoid.

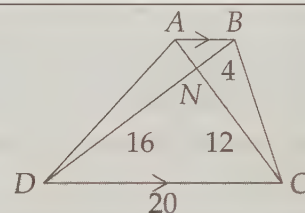
In this book, we will be careful to present proofs and definitions based on trapezoids as being valid for either of these definitions. In this section, we will focus on quadrilaterals with exactly one pair of parallel sides.

Problems

Problem 8.2: In trapezoid $ABCD$, we have $\overline{AB} \parallel \overline{CD}$, $\angle A = 150^\circ$, and $\angle B = 80^\circ$. Find $\angle C$ and $\angle D$.

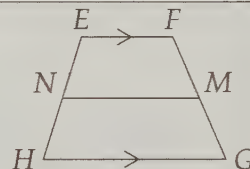


Problem 8.3: In trapezoid $ABCD$, $\overline{AB} \parallel \overline{CD}$, diagonals $\overline{BD}$ and $\overline{AC}$ meet at N , and side lengths BN , CN , DN , and CD are as shown in the diagram. Find AN and AB .



Problem 8.4: In trapezoid $EFGH$, we have $\overline{EF} \parallel \overline{GH}$, $EF = 6$, $FG = 10$, $GH = 12$, and $EH = 8$. Our goal is to find the length of median $\overline{MN}$.

- Extend $\overline{EH}$ and $\overline{FG}$ so that the two meet at X .
- Use similar triangles to find XE and XF , then find XN and XM .
- Show that $\triangle XEF \sim \triangle XNM$. Is the median parallel to the bases of the trapezoid?
- Find MN .



Problem 8.5: Bases $\overline{PQ}$ and $\overline{SR}$ of trapezoid $PQRS$ have lengths 5 and 9, respectively, and are 8 units apart.

- Find $[PQS]$ and $[QRS]$.
- Find $[PQRS]$.
- Can you find a general formula for the area of a trapezoid given its base lengths and the distance between the bases?

Problem 8.6: In trapezoid $WXYZ$, we have $\overline{WX} \parallel \overline{YZ}$, $WX = 17$, $XY = 13$, $YZ = 21$, and $XZ = 20$. In this problem we seek the area of $WXYZ$.

- We build right triangles by drawing an altitude from X to $\overleftrightarrow{YZ}$. How do we know this altitude meets $\overleftrightarrow{YZ}$ between Y and Z ?
- Let the length of the altitude we draw in part (a) be h , and let it split $\overline{YZ}$ into segments of lengths x and $21 - x$. Use either the Pythagorean Theorem and algebra or your knowledge of Pythagorean triples to find x and h .
- Find the area of $WXYZ$.

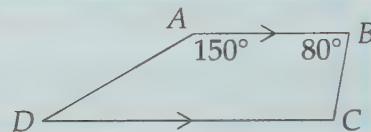
Problem 8.7: In trapezoid $ABCD$, we have $\overline{AD} \parallel \overline{BC}$, $\angle B = \angle C$, and $\angle B < 90^\circ$. Prove that $AB = CD$.
Hints: 131

Problem 8.8: In trapezoid $GHIJ$, $\overline{GH} \parallel \overline{IJ}$, $GH = 8$, $HI = GJ = 6$, and $IJ = 10$. In this problem we will find HJ .

- Draw heights $\overline{HX}$ and $\overline{GY}$. Find JY , YX , and XI .
- Find HX , then find HJ .

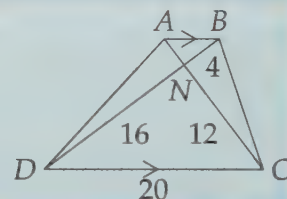
Many trapezoid problems are solved using the basic properties of parallel lines.

Problem 8.2: In trapezoid $ABCD$, we have $\overline{AB} \parallel \overline{CD}$, $\angle A = 150^\circ$, and $\angle B = 80^\circ$. Find $\angle C$ and $\angle D$.



Solution for Problem 8.2: Since $\overline{AB} \parallel \overline{CD}$, we have $\angle A + \angle D = 180^\circ$, so $\angle D = 30^\circ$. Similarly, $\angle C = 180^\circ - \angle B = 100^\circ$. $\square$

Problem 8.3: In trapezoid $ABCD$, $\overline{AB} \parallel \overline{CD}$, diagonals $\overline{BD}$ and $\overline{AC}$ meet at N , and lengths BN , CN , DN , and CD are as shown in the diagram. Find AN and AB .



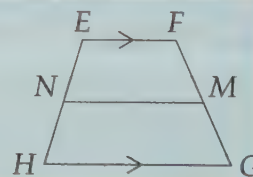
Solution for Problem 8.3: Parallel lines mean similar triangles: $\overline{AB} \parallel \overline{CD}$ gives $\triangle ABN \sim \triangle CDN$, so we have

$$\frac{AN}{CN} = \frac{BN}{DN} = \frac{AB}{CD}.$$

Since we have $BN/DN = 1/4$, we find $AN = CN/4 = 3$ and $AB = CD/4 = 5$. $\square$

Now we turn our attention to the median and the area of a trapezoid. The median of a trapezoid connects the midpoints of the legs of the trapezoid.

Problem 8.4: In trapezoid $EFGH$, we have $\overline{EF} \parallel \overline{GH}$, $EF = 6$, $FG = 10$, $GH = 12$, and $EH = 8$. Find the length of median $\overline{MN}$.



Solution for Problem 8.4: Make sure you understand why this solution is incomplete:

Bogus Solution: Since M is midway between the lines, and so is N , $\overline{MN}$ must be parallel to the bases and midway between the lengths of the bases. Therefore, MN is the average of the bases, so $MN = (EF + GH)/2 = 9$.



This solution has several problems. It consists of a series of statements without sufficient justification. For example, it doesn't explain what 'M is midway between the lines' means, nor why this leads to $\overline{MN}$ being parallel to the bases. We also have absolutely no justification for the claim that MN is the average of EF and GH .

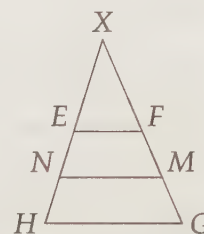
Many students new to writing mathematics write solutions that look much like this Bogus Solution. Strive to explain all your steps in mathematical terms, and to justify your claims. Showing exactly why MN is the average of EF and GH is not a trivial task. $\overline{MN}$ looks parallel to the bases, but we'll have to go through a few steps to prove it. We want to use triangles since we know a lot about triangles, so we extend $\overline{EH}$ and $\overline{FG}$ to meet at X . With this inspiration as a starting point, here's what a complete solution looks like:

The extensions of the legs meet at X . Since $\triangle EXF \sim \triangle HXG$, we have

$$\frac{XE}{XH} = \frac{EF}{GH}.$$

So, substitution gives

$$\frac{XE}{XE + 8} = \frac{1}{2}.$$



Solving this equation, we find $XE = 8$. Similarly, since $XF/XG = 1/2$, F is the midpoint of $\overline{XG}$, so $XF = FG = 10$. M is the midpoint of $\overline{FG}$, so $XM = XF + FG/2 = 15$. Similarly, we have $XN = 12$. Since $XN/XE = 12/8 = 3/2$ and $XM/XF = 15/10 = 3/2$, we have $\triangle XEF \sim \triangle XNM$ by SAS Similarity. Therefore, $NM/EF = XN/XE = 3/2$, so $NM = (3/2)(EF) = 9$. Notice also that $\angle XNM = \angle XEF$ from our similarity, so $\overline{NM} \parallel \overline{EF}$. $\square$

Important: The median of a trapezoid is parallel to the bases of a trapezoid, and equal in length to the average of the lengths of the bases.



You'll have a chance at the end of this section to use the same process we used on Problem 8.4 to prove these assertions for all trapezoids.

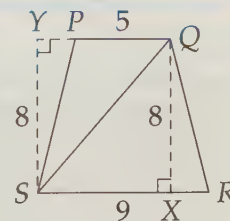
We used triangles to learn about the median; now we use them to learn about the area of a trapezoid.

Problem 8.5: Bases $\overline{PQ}$ and $\overline{SR}$ have lengths 5 and 9, respectively, and are 8 units apart. What is the area of trapezoid $PQRS$?

Solution for Problem 8.5: We know how to find the area of triangles, so we divide the trapezoid into two triangles by drawing diagonal $\overline{QS}$. We then draw the altitudes $\overline{QX}$ and $\overline{SY}$ of the two triangles. Since the parallel lines are 8 units apart, $QX = SY = 8$. We can now find the area of the trapezoid by adding the areas of the triangles:

$$[PQRS] = [PQS] + [QRS] = \frac{(PQ)(SY)}{2} + \frac{(SR)(QX)}{2} = 8 \left(\frac{PQ + SR}{2} \right) = 56.$$

$\square$



Since the median of a trapezoid is the average of the bases of the trapezoid, we can use our last problem to write a formula for the area of a trapezoid in terms of the height and median and a trapezoid.

Important: The area of a trapezoid equals the height of the trapezoid times the length of the median of the trapezoid.

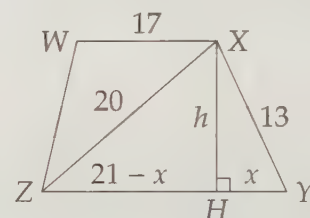


Problem 8.6: In trapezoid $WXYZ$, we have $\overline{WX} \parallel \overline{YZ}$, $WX = 17$, $XY = 13$, $YZ = 21$, and $XZ = 20$. Find the area of $WXYZ$.

Solution for Problem 8.6: We need the height, so we try to add a height to the diagram in a way that forms useful right triangles. The altitude from X to $\overleftrightarrow{YZ}$ is the most promising candidate, but we first have to figure out where the altitude from X will hit $\overleftrightarrow{YZ}$. It's not immediately clear if the foot of the altitude, which we'll call H , is on $\overline{YZ}$ or not. (It might be beyond Y on $\overleftrightarrow{ZY}$ for example.) Wherever H is on $\overleftrightarrow{YZ}$, $\triangle XZH$ is a right triangle. Since $\overline{XZ}$ is the hypotenuse and $\overline{ZH}$ a leg of this triangle, we have $HZ < XZ$. Since $XZ < YZ$, we know that $HZ < YZ$. Similarly, we have $HY < XY < YZ$. Since HZ and HY are both less than YZ , H must be on $\overline{YZ}$. Now we can confidently draw our diagram with H on $\overline{YZ}$.

We let $XH = h$ and $HY = x$, so that $ZH = 21 - x$ and we can now use the Pythagorean Theorem on $\triangle XHZ$ and $\triangle XHY$:

$$\begin{aligned} x^2 + h^2 &= 169 \\ (21 - x)^2 + h^2 &= 400 \end{aligned}$$



Subtracting the first equation from the second allows us to eliminate h and find x . We get

$$[(21 - x)^2 + h^2] - (x^2 + h^2) = 400 - 169.$$

Therefore, $x^2 - 42x + 441 + h^2 - x^2 - h^2 = 231$, so $-42x = -210$, which gives $x = 5$.

Substitution then gives us $h = 12$. Seeing such nice neat numbers, we realize that we could have guessed $h = 12$, since a leg of length 12 goes with hypotenuses $XZ = 20$ and $XY = 13$ nicely. (Notice that just because $h = 12$ fits these hypotenuses nicely doesn't mean that h must be 12. We still have to check that $ZY = 21$ if we guess $h = 12$.)

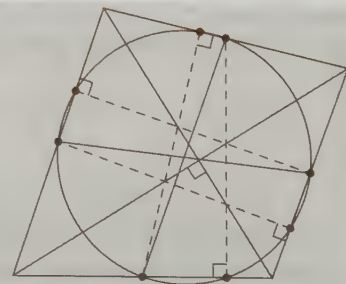
Now that we have $h = 12$, we can find $[WXYZ] = (h)(WX + YZ)/2 = (12)(38)/2 = 228$. $\square$

Concept: Dropping well-chosen altitudes to build right triangles is a powerful problem-solving tool in trapezoid problems.



Extra! Consider a quadrilateral with perpendicular diagonals as shown at right. The **Eight-Point Circle Theorem** tells us that the midpoints of the sides and the feet of the perpendiculars from these midpoints to the opposite respective sides of the quadrilateral all lie on the same circle.

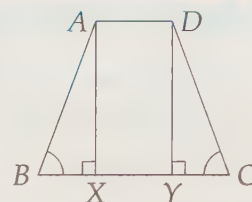
If you can't prove the Eight-Point Circle Theorem now, try coming back to it after you've tackled Problem 8.19. You might find the result of Problem 7.6 useful, as well.



We'll now take a look at a special class of trapezoids.

Problem 8.7: In trapezoid $ABCD$, we have $\overline{AD} \parallel \overline{BC}$, $\angle B = \angle C$, and $\angle B < 90^\circ$. Prove that $AB = CD$.

Solution for Problem 8.7: Once again, we introduce right triangles by drawing altitudes from A and D . Then $AX = DY$ because $\overline{AD} \parallel \overline{BC}$, so by AAS Congruence, we have $\triangle ABX \cong \triangle DCY$. Therefore, $AB = CD$.



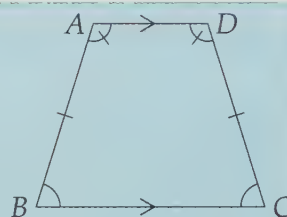
Notice that we can easily deal with the case of $\angle B$ and $\angle C$ being obtuse by noting that if $\angle B = \angle C$, then $\angle A = 180^\circ - \angle B = 180^\circ - \angle C = \angle D$. If $\angle B$ and $\angle C$ are obtuse, then $\angle A$ and $\angle D$ are acute, and we can use essentially the same proof as above to prove that the legs of the trapezoid are equal in length. $\square$

Two angles of a trapezoid that share one of the bases of the trapezoid as a side are together called **base angles** of the trapezoid. A trapezoid in which two base angles are equal is called an **isosceles trapezoid**. As an Exercise, you'll also prove that the diagonals of an isosceles trapezoid are congruent. Finally, in Section 8.7, you'll show that these relationships run 'backwards'; i.e., that if the legs are equal in length (and not parallel) or if the diagonals of a trapezoid are congruent, then both pairs of base angles are equal.

Important: In an isosceles trapezoid:



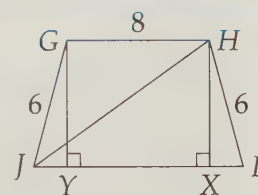
- (a) The base angles come in two pairs of equal angles as shown at right.
- (b) The legs are equal.
- (c) The diagonals are equal.



If any one of these items is true for a trapezoid with exactly one pair of parallel sides, then all the others must be true for that trapezoid.

Problem 8.8: Find HJ given that $GHIJ$ is an isosceles trapezoid with $GH = 8$, $HI = GJ = 6$, and $IJ = 10$.

Solution for Problem 8.8: We start by drawing altitudes $\overline{GY}$ and $\overline{HX}$, hoping to build useful right triangles. We note that since $GY = HX$ and $GJ = HI$, we have $\triangle GJY \cong \triangle HIX$ by HL Congruence. Therefore, $JY = XI$. Next, we see that $YX = GH$ because $\overline{GY} \parallel \overline{HX}$ and YX and GH are both the distance between $\overline{GY}$ and $\overline{HX}$. (Alternatively, we could note that $GHXY$ is a rectangle.) Therefore, $YX = 8$, so $JY = XI = (10 - 8)/2 = 1$.



Now we find HX from $\triangle HXI$: $HX = \sqrt{HI^2 - IX^2} = \sqrt{35}$. Finally, we use right triangle HXJ to get HJ : $HJ = \sqrt{HX^2 + XJ^2} = \sqrt{35 + 81} = \sqrt{116} = 2\sqrt{29}$. Building right triangles to find lengths strikes again. $\square$

If you found this section a little tough, review it after reading the rest of this chapter. You'll find that the other quadrilateral types are much simpler to handle than trapezoids, primarily because our other

types have restrictions on all the sides instead of just on one pair of opposite sides. Therefore, the other quadrilateral types don't offer as wide a variety of problems as trapezoids do.

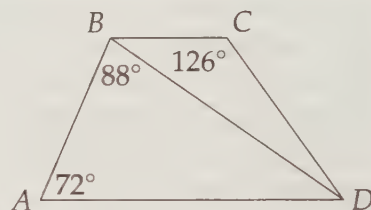
Exercises

8.2.1 In quadrilateral $PQRS$, $\overline{PQ} \parallel \overline{RS}$, $\overline{PS}$ is not parallel to $\overline{QR}$, $QR = PS$, and $\angle P = 83^\circ$. Find the rest of the angles of the trapezoid.

8.2.2 Find the area of a trapezoid with bases 44 and 24 and with height 18.

8.2.3 The area of trapezoid $ABCD$ is 96. One base is 6 units longer than the other, and the height of the trapezoid is 8. Find the length of the shorter base.

8.2.4 $ABCD$ shown at right is a trapezoid with $\overline{BC} \parallel \overline{AD}$, $\angle ABD = 88^\circ$, $\angle A = 72^\circ$, and $\angle C = 126^\circ$. Find $\angle ADC$ and $\angle CBD$.



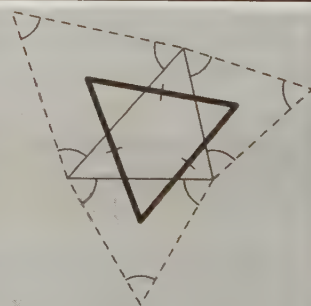
8.2.5 Prove that the diagonals of an isosceles trapezoid are congruent.

8.2.6 In this problem we prove that if $ABCD$ is a trapezoid such that $\overline{AB} \parallel \overline{CD}$ and $AB < CD$, then the median of $ABCD$ is parallel to the bases of $ABCD$ and equals the average of the bases of the trapezoid. Use Problem 8.4 as your guide. (Note that this proof covers any trapezoid in which the legs are not parallel.)

- Let M and N be the midpoints of $\overline{AD}$ and $\overline{BC}$ respectively. Extend $\overline{AD}$ and $\overline{BC}$ to meet at X . Prove $\triangle XDC \sim \triangle XAB$.
- Prove that $AD/XA = BC/XB$, and use this to prove that $AM/XA = BN/XB$.
- Prove that $XM/XA = XN/XB$.
- Prove that $\overline{MN} \parallel \overline{AB}$.
- ★ Prove that $MN = (AB + CD)/2$. **Hints:** 112, 153

8.2.7★ The bases of a trapezoid have lengths 50 and 75. Its diagonals have lengths 35 and 120. Find the area of the trapezoid. **Hints:** 339, 420

Extra! If we construct an equilateral triangle on the outside of each side of any triangle, the triangle formed by connecting the centroids of these equilateral triangles is called the **Outer Napoleon Triangle**. As the diagram strongly suggests, the outer Napoleon triangle is always equilateral. As you might guess, we can construct the equilateral triangles 'inward' and connect their centers to build an **Inner Napoleon Triangle**. Draw a triangle, then construct the Inner Napoleon Triangle. Does it look equilateral, too?



While the French emperor Napoleon Bonaparte is generally given credit for discovering these triangles, there's no solid proof that he actually did so.

8.3 Parallelograms

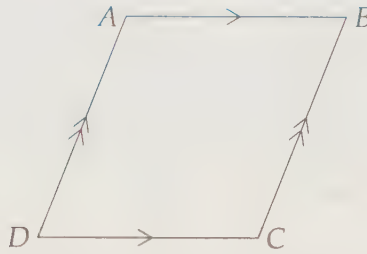
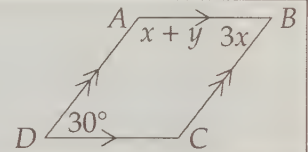


Figure 8.5: A Parallelogram

A quadrilateral is a **parallelogram** if both pairs of opposite sides are parallel. $ABCD$ in Figure 8.5 is a parallelogram because $\overline{AB} \parallel \overline{CD}$ and $\overline{AD} \parallel \overline{BC}$.

Problems

Problem 8.9: Given that $\angle A = x + y$, $\angle B = 3x$, and $\angle D = 30^\circ$ in parallelogram $ABCD$ as shown, find x , y , and $\angle C$.



Problem 8.10: In this problem we show that if $AB = CD$ and $AD = BC$, then $ABCD$ is a parallelogram.

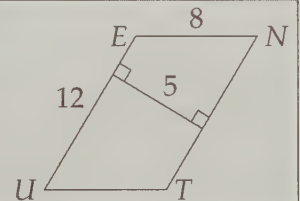
- Draw a parallelogram $ABCD$ with diagonal $\overline{AC}$. Prove that $\triangle BCA \cong \triangle DAC$.
- Prove that $\overline{AB} \parallel \overline{CD}$.
- Prove that $ABCD$ is a parallelogram.
- If $ABCD$ is a parallelogram, must $AD = BC$ and $AB = CD$? Make sure you see how this is a different question from the one we have already tackled.

Problem 8.11:

- Prove that if $EFGH$ is a parallelogram, then diagonals $\overline{EG}$ and $\overline{FH}$ bisect each other.
- Does this work in reverse? In other words, given that diagonals $\overline{WY}$ and $\overline{XZ}$ of quadrilateral $WXYZ$ bisect each other, is $WXYZ$ a parallelogram?

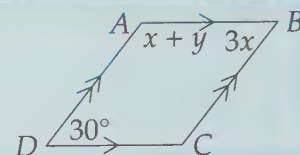
Problem 8.12: Sides $\overline{EU}$ and $\overline{NT}$ of parallelogram $ENTU$ are 5 units apart as shown. $EU = 12$ and $EN = 8$.

- Find the area of $ENTU$. (Hint: Remember the dissection we used to get the area of a trapezoid.)
- How far apart are sides $\overline{EN}$ and $\overline{TU}$?



As we'll see, parallelograms are a little easier to handle than quadrilaterals that have only one pair of parallel sides. Once again, our primary tools will be triangles and parallel lines.

Problem 8.9: Given that the measures of the angles of parallelogram $ABCD$ are as shown, find x , y , and $\angle C$.



Solution for Problem 8.9: Since $\overline{AD} \parallel \overline{BC}$, we have $\angle D + \angle C = 180^\circ$, so $\angle C = 150^\circ$. Similarly, $\overline{AB} \parallel \overline{CD}$ means $\angle B + \angle C = 180^\circ$, so $\angle B = 30^\circ$. Therefore, $3x = 30^\circ$, so $x = 10^\circ$. Finally, $\angle A + \angle D = 180^\circ$, so $\angle A = 150^\circ$. Hence, $x + y = 150^\circ$, so $y = 140^\circ$. $\square$

Notice that in our solution we found that $\angle B = \angle D$ and $\angle A = \angle C$. At the end of the section, you'll be asked to prove that the opposite angles of any parallelogram are equal.

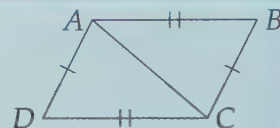
Problem 8.10: Show that if $AB = CD$ and $AD = BC$, then $ABCD$ is a parallelogram.

Solution for Problem 8.10: Where did we go wrong here:

Bogus Solution:



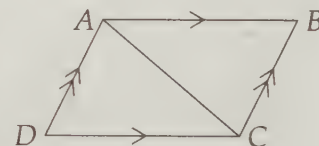
We cut $ABCD$ into triangles by drawing $\overline{AC}$. Since $\overline{AD} \parallel \overline{BC}$, we have $\angle DAC = \angle BCA$. Combining this with $AD = BC$ and $AC = AC$, we have $\triangle DAC \cong \triangle BCA$ by SAS Congruence. Therefore, $\angle BAC = \angle ACD$, so $\overline{AB} \parallel \overline{CD}$. Similarly, we can show that $\overline{AD} \parallel \overline{BC}$, so $ABCD$ is a parallelogram.



Our error is that we use $\overline{AD} \parallel \overline{BC}$ to prove that $\overline{AB} \parallel \overline{CD}$. However, we don't know that $\overline{AD} \parallel \overline{BC}$, so we can't use it yet. Fortunately, patching the solution is easy.

Still using our figure from the Bogus Solution above, we have $\triangle DAC \cong \triangle BCA$ by SSS Congruence. Therefore, $\angle BAC = \angle ACD$, so $\overline{AB} \parallel \overline{CD}$. Similarly, we have $\angle DAC = \angle ACB$, so $\overline{AD} \parallel \overline{BC}$ as well. Thus, $ABCD$ is a parallelogram. $\square$

We can also quickly show that in any parallelogram, the opposite sides must be equal. We do so by again using diagonal $\overline{AC}$ of parallelogram $ABCD$. We have $\triangle ABC \cong \triangle CDA$ by ASA Congruence. These congruent triangles give us $AB = CD$ and $BC = AD$. Make sure you see why this addresses a different question from the one initially asked in this problem!



Extra! In the spirit of David Hilbert's famous 1900 lecture (see page 56), in 2000 the Clay Mathematics Institute offered \$1,000,000 prizes for solutions to each of seven unsolved **Millennium Problems**. Russian mathematician **Grigori Perelman** put forth a solution to one of the Millennium problems, the **Poincaré Conjecture**, in 2003. His solution was so complex that it took roughly three years to validate! As of June, 2009, the Clay Mathematics Institute had not yet awarded the prize, since Perelman's proof has not been formally published in a peer-reviewed journal.

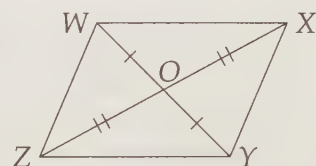
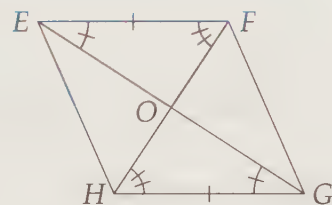
We've looked at the sides and the angles of a parallelogram; let's check out the diagonals.

Problem 8.11:

- Prove that if $EFGH$ is a parallelogram, then diagonals $\overline{EG}$ and $\overline{FH}$ bisect each other.
- Does this work in reverse? In other words, given that diagonals $\overline{WY}$ and $\overline{XZ}$ of quadrilateral bisect each other, is $WXYZ$ a parallelogram?

Solution for Problem 8.11:

- Since $EFGH$ is a parallelogram, we have $EF = GH$ and $\overline{EF} \parallel \overline{GH}$. Therefore, $\angle FEG = \angle EGH$ and $\angle EFH = \angle FGH$, so $\triangle EFO \cong \triangle GHO$ by ASA Congruence. Therefore, $EO = OG$ and $FO = OH$, so O is the midpoint of both diagonals of $EFGH$. Thus, the diagonals of a parallelogram bisect each other.
- Once again, we mark the information we're given – the fact that the diagonals bisect each other. We see that $\triangle WOZ \cong \triangle YOX$ by SAS Congruence, so $\angle ZWY = \angle WYX$, which means $\overline{WZ} \parallel \overline{XY}$. Similarly, $\triangle WOX \cong \triangle YOZ$, which we can use to show $\overline{WX} \parallel \overline{YZ}$. Hence, if the diagonals of a quadrilateral bisect each other, then the quadrilateral is a parallelogram.



□

We can combine the information from the last three problems to see parallelograms the way an experienced geometer sees them.

Important:

In parallelogram $ABCD$, the opposite sides are equal, the opposite angles are equal, and the diagonals bisect each other.

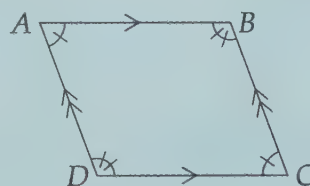
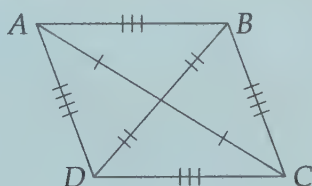


Figure 8.6: Side and Angle Equalities in Parallelograms

Conversely, $ABCD$ is a parallelogram if *any one* of the following are true:

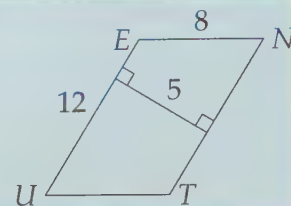
- $AB = CD$ and $AD = BC$.
- $\angle A = \angle C$ and $\angle B = \angle D$.
- Diagonals $\overline{AC}$ and $\overline{BD}$ bisect each other.

Proving one of these means the other two are true.

Finally, we tackle the area of a parallelogram. We could use the same tactic as we used with trapezoids, but there is an even easier approach.

Problem 8.12: Sides $\overline{EU}$ and $\overline{NT}$ of parallelogram $ENTU$ are 5 units apart as shown. $EU = 12$ and $EN = 8$.

- Find the area of parallelogram $ENTU$.
- How far apart are sides $\overline{EN}$ and $\overline{TU}$?



Solution for Problem 8.12:

- We can approach the area of a parallelogram the same way as we did with a trapezoid. If we draw diagonal $\overline{ET}$, we cut the parallelogram into two triangles, $\triangle EUT$ and $\triangle NET$, which have the same base ($EU = NT = 12$) and the same height, 5. Hence, we have

$$[ENTU] = [EUT] + [NET] = \frac{(EU)(5)}{2} + \frac{(NT)(5)}{2} = \frac{(EU)(5)}{2} + \frac{(EU)(5)}{2} = (EU)(5) = 60.$$

Notice that we can easily use this approach to show that the area of a parallelogram is its base times its height to that base.

- We have the base, EN , and the area from the previous part. Since the area of $ENTU$ is 60, the height between $\overline{EN}$ and $\overline{TU}$ is

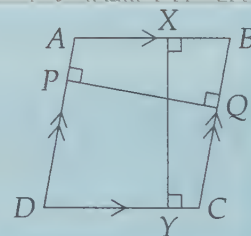
$$\frac{[ENTU]}{EN} = \frac{60}{8} = \frac{15}{2}.$$

□

Important: The area of a parallelogram is its base times its height.



$$[ABCD] = (AB)(XY) = (BC)(PQ).$$



Exercises

8.3.1 $TYUI$ is a parallelogram with $TY = 6$ and $YU = 8$.

- What is the perimeter of $TYUI$?
- Do we have enough information to find the area of $TYUI$?

8.3.2 $WXYZ$ is a parallelogram. Prove that $\angle W = \angle Y$.

8.3.3 In quadrilateral $WORK$, $WO = RK$ and $\overline{WO} \parallel \overline{RK}$. Prove that $WORK$ is a parallelogram.

8.3.4 Use a clever dissection of a parallelogram to turn it into a rectangle and prove that the area of the parallelogram is its base times its height. **Hints:** 394

8.3.5 Is it possible for a parallelogram $ENTU$ to have $EN = 4$, $EU = 12$, and for sides $\overline{EU}$ and $\overline{NT}$ to be 5 units apart? Why or why not? **Hints:** 528

8.3.6 In this Exercise, we give another way of deriving the length of the median and area of a trapezoid.

Let $ABCD$ be a trapezoid with bases $\overline{AB}$ and $\overline{CD}$.

- Extend $\overline{AB}$ past B to A' such that $BA' = CD$, and extend $\overline{DC}$ past C to D' such that $CD' = AB$. Prove that $AA'D'D$ is a parallelogram.
- Find the area of $[AA'D'D]$ in terms of AB , CD , and h , the distance between $\overline{AB}$ and $\overline{CD}$. Find the relationship between $[AA'D'D]$ and $[ABCD]$, and then find $[ABCD]$ itself. **Hints:** 341
- Let M , N , and M' be the midpoints of $\overline{AD}$, $\overline{BC}$, and $\overline{A'D'}$, respectively. Prove that $MM' = AB + CD$, and prove that $MN = MM'/2$. Conclude that $MN = (AB + CD)/2$.

8.3.7★ The diagonals of convex quadrilateral $ABCD$ meet at E . Prove that the centers of the circumcircles of $\triangle ABE$, $\triangle BCE$, $\triangle CDE$, and $\triangle DAE$ are the vertices of a parallelogram. **Hints:** 80

8.4 Rhombi

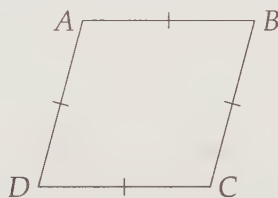


Figure 8.7: A Rhombus

A quadrilateral is a **rhombus** if all of its sides are equal.

Problems

Problem 8.13: Prove that every rhombus is a parallelogram.

Problem 8.14: $ABCD$ is a rhombus such that its diagonals $\overline{AC}$ and $\overline{BD}$ have lengths $AC = 30$ and $BD = 16$. In this problem, we will find the area of $ABCD$ and the side length of the rhombus.

- Let the diagonals intersect at X . Show that $\triangle AXB$, $\triangle CXB$, $\triangle CXD$, and $\triangle AXD$ are congruent.
- Prove that $\overline{AC} \perp \overline{BD}$.
- Find the area of $ABCD$.
- Find AB .

We can prove a whole lot about rhombi in one simple step.

Problem 8.13: Prove that every rhombus is a parallelogram.

Solution for Problem 8.13: Let $ABCD$ be a rhombus. Since $AB = CD$ and $AD = BC$ (because all the sides of $ABCD$ are equal), we know that $ABCD$ is a parallelogram. $\square$

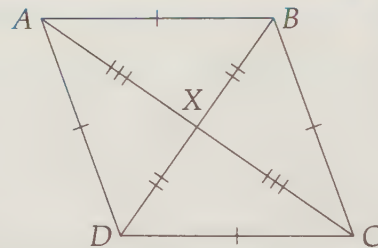
Important: Every rhombus is a parallelogram. Therefore, all that is true about parallelograms is true about rhombi.

WARNING!! (Bet you saw this one coming.) Every parallelogram is *not* a rhombus. Therefore, if we prove a property of rhombi, this property is not necessarily true for all parallelograms – we’d have to prove the property for parallelograms separately.

To see some of the properties that distinguish a rhombus from a parallelogram that is not a rhombus, let’s try a problem.

Problem 8.14: $ABCD$ is a rhombus such that its diagonals $\overline{AC}$ and $\overline{BD}$ have lengths $AC = 30$ and $BD = 16$. Find AB and $[ABCD]$.

Solution for Problem 8.14: We start with a diagram, in which we note that all the sides of $ABCD$ are equal, and that the diagonals bisect each other since rhombus $ABCD$ is also a parallelogram. All four little triangles in the diagram are congruent by SSS Congruence. Since all four angles at X must be equal and they add to 360° , they must be right angles. Hence, the diagonals of a rhombus not only bisect each other, but they are also perpendicular. Since each of these right triangles have legs of length 8 and 15, their hypotenuses (which are the sides of the rhombus) each have length 17.



All those congruent right triangles give us a quick approach to finding the area.

$$[ABCD] = 4[ABX] = 4 \left(\frac{(AX)(BX)}{2} \right) = 2(AX)(BX) = 240.$$

Notice that since AX and BX are each half a diagonal of $ABCD$, we have

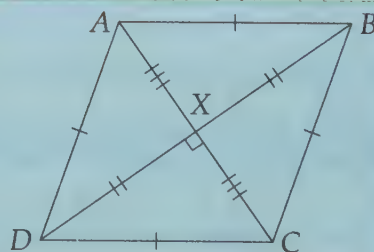
$$[ABCD] = 2(AX)(BX) = 2 \left(\frac{AC}{2} \right) \left(\frac{BD}{2} \right) = \frac{(AC)(BD)}{2}.$$

□

Our solution gives us a few special rhombus facts to add to all the facts we know about rhombi because they are parallelograms.

Important: The diagonals of a rhombus are perpendicular. The area of a rhombus is half the product of its diagonals (of course, it is also base times height since a rhombus is a parallelogram).

$$[ABCD] = \frac{(AC)(BD)}{2}$$



Exercises

- 8.4.1 $PQRS$ is a rhombus with diagonals $PR = 6$ and $QS = 12$. Find the area and the perimeter of $PQRS$.
- 8.4.2 $WXYZ$ is a rhombus with $WX = 50$ and $WY = 96$.
- Find XZ .
 - Find $[WXYZ]$.
 - Find the distance between $\overline{WX}$ and $\overline{YZ}$. **Hints:** 586
- 8.4.3 Diagonals AC and BD of quadrilateral $ABCD$ are perpendicular. Prove that $[ABCD] = (AC)(BD)/2$. **Hints:** 438
- 8.4.4 $TUVW$ is a rhombus with $TU = 10$ and $\angle TUV = 60^\circ$.
- Show that $\angle TUW = 30^\circ$.
 - Find $[TUVW]$.

8.5 Rectangles

We've looked at quadrilaterals in which all the sides are equal. Now we consider what happens if all the angles are equal. Such a quadrilateral is called a **rectangle**.

Problems

Problem 8.15: What are the measures of the angles in a rectangle?

Problem 8.16: Prove that all rectangles are also parallelograms.

Problem 8.17: Prove that the diagonals of a rectangle are congruent.

Problem 8.18: Find the length of a diagonal of a rectangle given that its perimeter is 44 and one side has length 10.

Problem 8.19: In this problem, we find two different proofs that the quadrilateral formed by connecting the midpoints of a rhombus is a rectangle.

- Start with a diagram. Draw rhombus $ABCD$ and let $EFGH$ be the quadrilateral formed by connecting the midpoints of its sides, with E on $\overline{AB}$, F on $\overline{BC}$, and so on.
- Solution 1:* Show that $\angle AEH = \angle AHE$ and that $\triangle AEH \cong \triangle CGF$. Use this as a starting point to show that the angles of $EFGH$ are all equal.
- Solution 2:* Show that $\overline{EF} \parallel \overline{AC}$. Show that the angles of $EFGH$ are right angles.

We know the angles of a rectangle are equal, but to what?

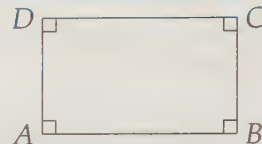
Problem 8.15: What are the measures of the angles in a rectangle?

Solution for Problem 8.15: The angles must add to 360° and they're all equal, so each of the four angles must be a right angle. $\square$

As we did with rhombi, we can quickly get a wealth of information about rectangles.

Problem 8.16: Prove that all rectangles are also parallelograms.

Solution for Problem 8.16: Let $ABCD$ be our rectangle. Since $\angle A + \angle B = 180^\circ$, we know that $\overline{AD} \parallel \overline{BC}$. Similarly, $\angle B + \angle C = 180^\circ$, so $\overline{AB} \parallel \overline{CD}$. Therefore, $ABCD$ is a parallelogram. $\square$

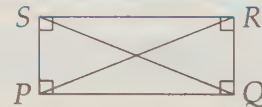


Among the many useful aspects of knowing that every rectangle is a parallelogram, the one that is probably most used is the fact that opposite sides are equal in length. When talking about rectangles, these two dimensions are often called the *length* and *width* of the rectangle. We have already seen that the area of a rectangle is the product of these two dimensions.

We know that the diagonals of a rhombus are perpendicular, so we wonder if there's a similar special property of the diagonals of a rectangle.

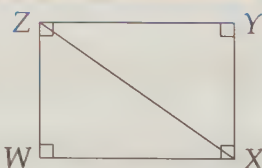
Problem 8.17: Prove that the diagonals of a rectangle are congruent.

Solution for Problem 8.17: As we have in many other problems, we turn to congruent triangles for a solution. Drawing our two diagonals, we have $\triangle PQR \cong \triangle QPS$ by SAS since $PS = QR$, $PQ = PQ$, and $\angle PQR = \angle QPS$. Therefore, $PR = QS$. $\square$



Problem 8.18: Find the length of a diagonal of a rectangle given that its perimeter is 44 and one side has length 10.

Solution for Problem 8.18: Let $WXYZ$ be our rectangle and let $XY = 10$. We know the diagonals have the same length, so we don't have to worry about which diagonal to find. Since $XY = 10$, we have $WZ = 10$ as well. Since the perimeter is 44, we have



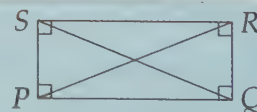
$$WX + XY + YZ + ZW = 44.$$

Since $WX = YZ$ and $XY = WZ = 10$, we have $WX + 10 + WX + 10 = 44$, so $WX = 12$. Now we can use the Pythagorean Theorem to find $XZ = \sqrt{WX^2 + WZ^2} = \sqrt{100 + 144} = 2\sqrt{61}$. $\square$

Important:



The diagonals of a rectangle are equal to each other and equal to the square root of the sum of the squares of the length and the width of the rectangle.



$$QS = PR = \sqrt{QR^2 + SR^2}$$

You shouldn't have to memorize how to find the diagonals of a rectangle: the right angles of a

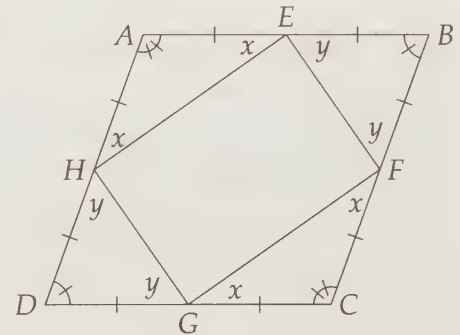
rectangle should make it clear to use the Pythagorean Theorem with the sides to get the diagonal.

Now we'll take a look at one way rectangles and rhombi are related, before moving on to a quadrilateral that is both a rhombus *and* a rectangle.

Problem 8.19: Prove that the quadrilateral formed by connecting the midpoints of the sides of a rhombus is a rectangle.

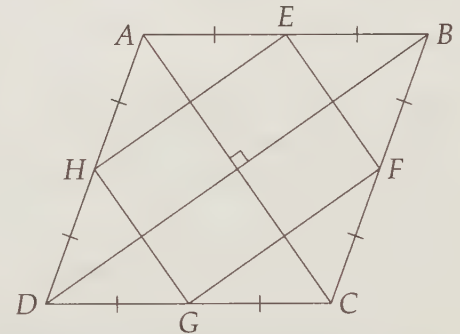
Solution for Problem 8.19: We draw rhombus $ABCD$ and let $EFGH$ be the quadrilateral formed by connecting the midpoints of the sides of $ABCD$, with E on $\overline{AB}$, F on $\overline{BC}$, and so on. We will take two different approaches to solving the problem.

- (a) Since we want to show that $EFGH$ is a rectangle, we look for a way to find the measures of each angle of $EFGH$. Since its vertices are the midpoints of $ABCD$ and $ABCD$ is a rhombus, the vertices of $EFGH$ divide the sides of $ABCD$ into 8 equal segments as shown.



Since $ABCD$ is a parallelogram, $\angle A = \angle C$ and $\angle B = \angle D$. Therefore, $\triangle AEH \cong \triangle CGF$ and $\triangle BEF \cong \triangle DGH$. Furthermore, each of these small triangles is isosceles, so we can let $\angle AEH = x$ and $\angle DGH = y$ and identify equal angles as shown. Finally, we see that each angle of $EFGH$ has measure $180^\circ - x - y$. Thus, the angles of $EFGH$ are equal and $EFGH$ must be a rectangle.

- (b) For another approach, we might note that we want to prove that the angles of $EFGH$ are right angles. This gets us thinking about where we can find right angles in $ABCD$. Since $ABCD$ is a rhombus, its diagonals are perpendicular. To relate the sides of the rectangle to these diagonals, we find that $\triangle AEH \sim \triangle ABD$ by SAS Similarity since $AE/AB = AH/AD = 1/2$ and $\angle EAH = \angle BAD$. Therefore, $\angle AEH = \angle ABD$, so $\overline{EH} \parallel \overline{BD}$. Similarly, we can show that $\overline{FG} \parallel \overline{BD}$. Since $\overline{AC}$ is perpendicular to $\overline{BD}$, it must be perpendicular to $\overline{EH}$ and $\overline{FG}$ as well, because both $\overline{EH}$ and $\overline{FG}$ are parallel to $\overline{BD}$.



We can use exactly the same approach to show that $\overline{EF} \parallel \overline{AC} \parallel \overline{GH}$, from which we can conclude that $\overline{EF}$ and $\overline{GH}$ are perpendicular to $\overline{BD}$. Now we can tie all these parallel and perpendicular lines together. Since $\overline{EF} \perp \overline{BD}$ and $\overline{FG} \parallel \overline{BD}$, we have $\overline{EF} \perp \overline{FG}$. In exactly the same way, we can show that each of the other three angles of $EFGH$ is a right angle. Thus, $EFGH$ is a rectangle.

□

Exercises

8.5.1 $POST$ is a rectangle with $PO = 8$ and $OS = 12$.

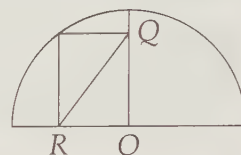
- Find the perimeter of $POST$.
- Find PS .

(c) Find $[POST]$.

8.5.2 The length of a rectangle is one less than twice its width. If the perimeter of the rectangle is 36, what is the area of the rectangle?

8.5.3 Diagonals $\overline{WR}$ and $\overline{ET}$ of rectangle $WERT$ meet at Y . Given that $\angle WYE = x$, find $\angle ERY$ and $\angle YRT$ in terms of x .

8.5.4 A semicircle with center O has a radius of 9 cm. What is the number of centimeters in the length of $\overline{RQ}$, a diagonal of the rectangle shown? (Source: MATHCOUNTS)



8.5.5 I have a 36 inch by 24 inch rectangular painting. I would like to place a frame that is 2 inches wide around the painting. If the material for the frame costs \$1.50 per square inch, how much will the frame cost?

8.5.6 What kind of quadrilateral do we get when we connect the midpoints of the sides of a rectangle? (Prove your answer!)

8.5.7★ $EFGH$ is a rectangle with area 48. If $EGJI$ is a rectangle such that H is on $\overline{JI}$, what is the area of $EGJI$? Hints: 434

8.6 Squares

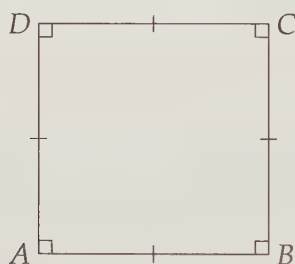


Figure 8.8: A Square

We've studied quadrilaterals with all their sides equal, and quadrilaterals with all their angles equal. What if a quadrilateral has both all sides equal and all angles equal? Such a quadrilateral is called a **square**, an example of which is shown in Figure 8.8.

As we'll see, squares are the easiest quadrilaterals to work with since everything that is true about rectangles, rhombi, and parallelograms is also true about squares.

Problems

Problem 8.20: Find the perimeter and area of a square with side length 9.

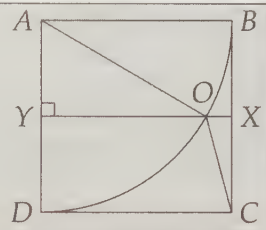
Extra! *The composer opens the cage door for arithmetic, the draftsman gives geometry its freedom.*



—Arthur Cayley

Problem 8.21: Find the perimeter and area of a square that has a diagonal of length 8.

Problem 8.22: $ABCD$ is a square. The circle with center A and radius AB meets the perpendicular bisector of $\overline{AD}$ in two points, of which O is the one inside the square. Find $\angle AOC$. **Hints:** 237



We've already studied all the tools we need for squares. How we use these tools in the following two problems should be unsurprising.

Problem 8.20: Find the perimeter and area of a square with side length 9.

Solution for Problem 8.20: Since all the sides of the square are equal, the perimeter of our square is $4 \times 9 = 36$. Since a square is a rectangle, its area is its length times its width. These dimensions are the same in a square, so the area is just the, um, 'square' of its side length, or $9^2 = 81$. (Now we see why multiplying a number by itself is called 'squaring' the number.) $\square$

Problem 8.21: Find the perimeter and area of a square that has a diagonal of length 8.

Solution for Problem 8.21: A square is a rectangle, so its diagonals are congruent. It is also a rhombus, so its area is half the product of its diagonals, or $(8^2)/2 = 32$ in this case.

We can find a side of the square by noting that a diagonal of a square splits the square into two 45-45-90 triangles, so each side equals the diagonal divided by $\sqrt{2}$, or $8/\sqrt{2} = 4\sqrt{2}$. Alternatively, since the area is just the square of the side length, we can take the square root of the area we already found: $\sqrt{32} = 4\sqrt{2}$. Thus, the perimeter is $4(4\sqrt{2}) = 16\sqrt{2}$. $\square$

These two examples give us the most useful length and area relationships for a square.

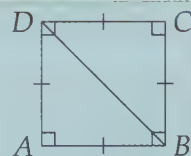
Important:



Let s be the side length of a square, P be the perimeter, and A the area. We have

$$P = 4s$$

$$A = s^2.$$



Drawing a diagonal creates two 45-45-90 triangles. Letting the length of a diagonal be d , we have

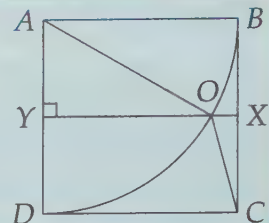
$$d = s\sqrt{2}$$

$$P = d(2\sqrt{2})$$

$$A = \frac{d^2}{2}.$$

We often encounter squares in problems together with circles, since both involve segments with equal lengths (sides for the square, radii for the circle).

Problem 8.22: $ABCD$ is a square. The circle with center A and radius AB meets the perpendicular bisector of $\overline{AD}$ in two points, of which O is the one inside the square. Find $\angle AOC$.



Solution for Problem 8.22: There doesn't seem any easy way to get $\angle AOC$, so we try splitting it into pieces by drawing $\overline{OD}$. Since $\triangle AYO \cong \triangle DYO$ by SAS, we have $AO = OD$. Furthermore, $AD = AO$ because they are radii of the circle. Thus, $\triangle AOD$ is equilateral, which implies $\angle AOD = 60^\circ$. Now we only have to find $\angle DOC$.

Since OD equals the side length of the square, we have $OD = DC$. Therefore, $\triangle DOC$ is isosceles, so $\angle DOC = \angle DCO$. Furthermore, we already know

$$\angle ODC = \angle ADC - \angle ADO = 90^\circ - 60^\circ = 30^\circ.$$

From $\triangle DOC$ we have

$$\angle DOC + \angle DCO + \angle ODC = 180^\circ,$$

so $2\angle DOC + 30^\circ = 150^\circ$. Therefore, $\angle DOC = 75^\circ$. Finally, we have $\angle AOC = \angle AOD + \angle DOC = 135^\circ$.

Note that you might also have found $\angle AOD = 60^\circ$ by seeing that $AO = AD = 2AY$ and $\angle AYO = 90^\circ$, so $\triangle AOY$ is a 30-60-90 triangle (and likewise for $\triangle DOY$.) $\square$

Concept: Breaking a desired angle, length, or area into parts often makes it easier to find.

Exercises

- 8.6.1 The area of square $EFGH$ is 80. Find EF and EG .
- 8.6.2 M is the midpoint of $\overline{AB}$ on square $ABCD$. If $\overline{AC}$ and $\overline{BD}$ meet at O , and $MO = 4$, what is the area of square $ABCD$?
- 8.6.3 The diagonals of square $TYUI$ meet at M . Point K is on side $\overline{TY}$ such that $TK = TM$. Find $\angle MTK$ and $\angle TMK$.
- 8.6.4 Show that a rectangle with perpendicular diagonals must be a square.
- 8.6.5 $ABCD$ and $ACFG$ are squares. Find $[ACFG]/[ABCD]$. **Hints:** 236
- 8.6.6 Point E is inside square $ABCD$ such that $\triangle ABE$ is equilateral. Given that $AB = 4$, find the following:
- (a) AE .
 - (b) $[ABCD]$.
 - (c) $[ABE]$.
 - (d) $\angle DAE$ and $\angle DEA$.
 - (e) The area inside $ABCD$ but outside $\triangle ABE$.
 - (f) CE .

8.7 If and Only If

In this section, we learn what ‘if and only if’ means by exploring what facts we need to classify a quadrilateral, and what facts we know given a quadrilateral’s classification.

We already have a good idea of what ‘if’ means. For example, we can write ‘An animal has four legs if it is a dog.’ However, ‘only if’ is a little trickier – for example, the statement ‘An animal has four legs only if it is a dog’ is clearly false. Cats have four legs, too.

Many mathematical statements put ‘if’ and ‘only if’ together. Here’s what it looks like in a non-mathematical context:

A month has less than thirty days if and only if the month is a February.

This statement is equivalent to saying both of the following statements at the same time:

Every month with less than thirty days is a February.

Every February has less than thirty days.

In other words, to prove an ‘if and only if’ statement, we must prove two different things. In our example above, we would have to show that every month that has less than thirty days is a February, and we would have to show that every February has less than thirty days. Make sure you understand that these are two different statements to prove!

Important: Proving ‘if and only if’ statements requires proving two different statements.



Problems

Problem 8.23: Prove that $ABCD$ is a parallelogram if and only if $\angle A = \angle C$ and $\angle B = \angle D$.

Problem 8.24:

- Prove that if $ABCD$ is a square, then its diagonals bisect its angles.
- Is it true to say that $ABCD$ is a square if and only if its diagonals bisect its angles?

Extra! *In the course of my law reading I constantly came upon the word demonstrate. I thought at first that I understood its meaning, but soon became satisfied that I did not. . . I consulted all the dictionaries and works of reference I could find. . . At last I said, ‘Lincoln, you can never make a lawyer if you do not understand what demonstrate means,’ and I left my situation in Springfield, went home to my father’s house and stayed there till I could give any proposition in the six books of Euclid at sight. I then found out what demonstrate means and went back to my law studies.*

—Abraham Lincoln

Problem 8.25: Prove that the diagonals of a trapezoid are congruent if and only if the trapezoid is isosceles. **Hints:** 216

Problem 8.26: In this problem, we prove that the circumcenter of a triangle is the same point as the incenter if and only if the triangle is equilateral.

- First, we tackle the 'if' part: Show that in an equilateral triangle, the circumcenter and the incenter are the same point.
- In the next three parts, we tackle the 'only if' part of the problem by showing that if the incenter and the circumcenter are the same point, then the triangle is equilateral. Let X be this common point and $\triangle ABC$ be our triangle. What does X being the circumcenter tell us about X ?
- What does X being the incenter tell us about X ?
- Combine your answers from the previous two parts to find congruent triangles. Use these triangles to deduce that $\triangle ABC$ is equilateral.

Now we'll try our hand at a few proofs (or disproofs) of 'if and only if' statements.

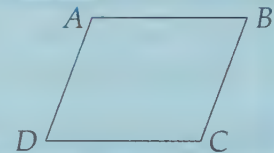
Problem 8.23: Prove that $ABCD$ is a parallelogram if and only if $\angle A = \angle C$ and $\angle B = \angle D$.

Solution for Problem 8.23: What's missing from this solution:

Bogus Solution:



Since $ABCD$ is a parallelogram, we have $\overline{AD} \parallel \overline{BC}$, so $\angle A + \angle B = 180^\circ$. Similarly, $\overline{AB} \parallel \overline{CD}$ means $\angle B + \angle C = 180^\circ$. Therefore, $\angle A = \angle C$. Similarly, we can show $\angle B = \angle D$.



This solution only does half the problem. 'If and only if' problems are two-part problems! We must prove two statements. First, we show:

$ABCD$ is a parallelogram if $\angle A = \angle C$ and $\angle B = \angle D$.

Since the angles of a quadrilateral add to 360° , we have $\angle A + \angle B + \angle C + \angle D = 360^\circ$. Since $\angle C = \angle A$ and $\angle D = \angle B$, we have $2(\angle A + \angle B) = 360^\circ$, so $\angle A + \angle B = 180^\circ$. Therefore, $\overline{AD} \parallel \overline{BC}$. Since $\angle A = \angle C$, we have $\angle B + \angle C = 180^\circ$ also. Therefore, $\overline{AB} \parallel \overline{CD}$ and $ABCD$ is a parallelogram.

Next, we must show:

$ABCD$ is a parallelogram only if $\angle A = \angle C$ and $\angle B = \angle D$.

In other words, we must show that every parallelogram $ABCD$ has $\angle A = \angle C$ and $\angle B = \angle D$. This is what we did in our Bogus Solution above, so we have already tackled this part.

Therefore, $ABCD$ is a parallelogram if and only if $\angle A = \angle C$ and $\angle B = \angle D$. $\square$

Sometimes one part of 'if and only if' is clearly true, but the other half isn't so obvious. And sometimes the other half isn't even true!

Problem 8.24:

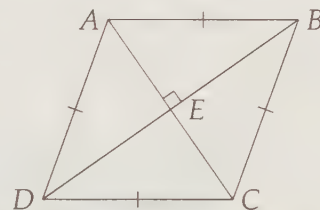
- (a) Prove that if $ABCD$ is a square, then its diagonals bisect its angles.
- (b) Is it true to say that $ABCD$ is a square if and only if its diagonals bisect its angles?

Solution for Problem 8.24:

- (a) We know that each diagonal of a square cuts the square into two isosceles right triangles. In other words, each diagonal cuts the right angles at the vertices into two 45° angles. So, if $ABCD$ is a square, its diagonals bisect its angles.
- (b) We've already proved one direction of this statement. The other direction is

Prove that $ABCD$ is a square if its diagonals bisect its angles.

Unfortunately, try as we may, we can't prove this! For example, the diagonals of any rhombus bisect the angles of the rhombus. We can show this quickly by noting that $\triangle AED \cong \triangle CED$ in rhombus $ABCD$ at right. Hence, $\angle ADE = \angle CDE$ and diagonal $\overline{BD}$ bisects $\angle ADC$. Similarly, each of the four angles of rhombus $ABCD$ is bisected by a diagonal of $ABCD$. Therefore, if the diagonals of a quadrilateral bisect its angles, we *cannot* deduce that the quadrilateral is a square, since the quadrilateral may be a rhombus that is not a square.



□

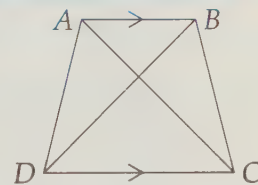
Important: To prove a statement is true, you must have a proof that covers all possibilities. To disprove a statement, all you have to do is find one case in which the statement is false. It's much easier to break something than to build it!



Let's try a couple more challenging 'if and only if' statements.

Problem 8.25: Prove that the diagonals of a trapezoid are congruent if and only if the trapezoid is isosceles.

Solution for Problem 8.25: Our diagram shows trapezoid $ABCD$ with $\overline{AB} \parallel \overline{CD}$. We'll assume that $\angle C < 90^\circ$, because if $\angle C > 90^\circ$, then we note $\angle B < 90^\circ$ and we can then use essentially the same proof as for $\angle C < 90^\circ$. (What happens if $\angle C = 90^\circ$?) Since we have an if and only if statement, we must prove two items. First, we must show:

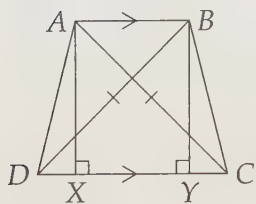


The diagonals of a trapezoid are congruent if the trapezoid is isosceles.

We must show that $AC = BD$ if $\angle ADC = \angle BCD$. Back in Problem 8.7, we showed that $\angle ADC = \angle BCD$ implies that $AD = BC$. Since we also have $\angle ADC = \angle BCD$ and $CD = CD$, we have $\triangle ADC \cong \triangle BCD$ by SAS. Therefore, $AC = BD$.

Next, we must show:

The diagonals of a trapezoid are congruent only if the trapezoid is isosceles.



In other words, we must show that if the diagonals are equal, the trapezoid is isosceles. So, we must start from $AC = BD$ and show that $\angle ADC = \angle BCD$. We would like to show that $\triangle ACD \cong \triangle BDC$, so we draw altitudes $\overline{AX}$ and $\overline{BY}$ as shown. Since $AC = BD$ and $AX = BY$ (because $\overline{AB} \parallel \overline{CD}$), we have $\triangle AXC \cong \triangle BYD$ by HL Congruence. Hence, we have $\angle ACD = \angle BDC$. Since we also have $AC = BD$ and $CD = CD$, we have $\triangle ACD \cong \triangle BDC$ by SAS Congruence.

Therefore, $\angle ADC = \angle BCD$.

Therefore, the diagonals of a trapezoid are congruent if and only if the trapezoid is isosceles. $\square$

You might be wondering, 'How in the world would we think to draw the altitudes we drew in the second part?' We think to add these because we have to somehow use the fact that $\overline{AB} \parallel \overline{CD}$. We could also have used this fact by finding some equal angles (this path is a little longer), but we know that drawing altitudes often simplifies trapezoid problems.

Concept: When stuck on a problem, think 'What information haven't I used yet?' Then, try to find some way to use that information.



Enough with quadrilaterals; let's go back to triangles.

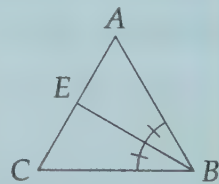
Problem 8.26: Prove that the circumcenter of a triangle is the same point as the incenter if and only if the triangle is equilateral.

Solution for Problem 8.26: What's wrong with the following solution:

Bogus Solution: Let ABC be our triangle. We can easily take care of showing that the circumcenter and incenter are the same point in an equilateral triangle since the angle bisectors of an equilateral triangle are the same lines as the perpendicular bisectors of the sides of the triangle.



Going the other way is tougher. Let $\overline{BE}$ be the angle bisector of $\angle B$. Since $AB = BC$, $\angle ABE = \angle CBE$, and $BE = BE$, we have $\triangle EBC \cong \triangle EBA$. So, $AE = EC$ and $\angle AEB = \angle CEB = 90^\circ$. Thus, angle bisector $\overline{BE}$ is part of the perpendicular bisector of $\overline{AC}$. Similarly, we can show that all the angle bisectors of $\triangle ABC$ are the same as the perpendicular bisectors, so the triangle must be equilateral.



This 'solution' has several missteps. First, we should be more explicit in the first part about showing why the angle bisectors and perpendicular bisectors are the same lines in an equilateral triangle. Second, and far more serious, is that our second part of the proof is logically flawed. We assume as a step in our 'proof' that $AB = BC$; however, we can't assume this because we don't know anything about the sides of $\triangle ABC$ in the second part yet – we are trying to prove that these are equal!

WARNING!! Be very careful that you do not assume what you are trying to prove as part of a proof!

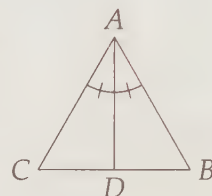


Finally, our 'proof' ends by stating that since the angle bisectors and the perpendicular bisectors are the same, the triangle must be equilateral. This statement, while true, is not proved in our solution – it's what we are asked to prove!

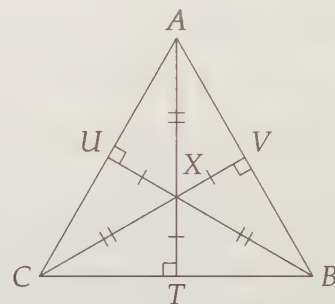
Let's try again:

Let $\triangle ABC$ be our triangle, O be the circumcenter, and I be the incenter.

First, we show that if a triangle is equilateral, then I and O are the same. We draw angle bisector $\overline{AD}$. Since $\triangle ABC$ is equilateral, we have $\triangle CAD \cong \triangle BAD$ by SAS. So, $CD = DB$ and $\angle CDA = \angle BDA = 90^\circ$. Therefore, $\overleftrightarrow{AD}$ is the perpendicular bisector of $\overline{BC}$. Similarly, each angle bisector of $\triangle ABC$ is also a perpendicular bisector of one of the sides. Thus, the intersection of the angle bisectors, the incenter, is the same point as the intersection of the perpendicular bisectors, the circumcenter.



Next, we turn to the second part, proving that if the incenter and the circumcenter are the same point, then the triangle must be equilateral. We let this common point be X . Since X is the incenter, it is equidistant from all three sides. Since it is the circumcenter, it is equidistant from all three vertices. These two sets of length equalities are shown in the diagram. We quickly see that all six little right triangles are congruent by HL Congruence. Hence, we have $AV = VB = BT = TC = CU = UA$, so $AB = BC = CA$ and the triangle is equilateral. (Note: There are plenty of other ways to do this part; perhaps you can find another way!) $\square$



Another phrase mathematicians will sometimes use to mean 'if and only if' is 'necessary and sufficient.' For example:

It is necessary and sufficient for a quadrilateral's diagonals to bisect each other in order for the quadrilateral to be a parallelogram.

This is the same as saying:

The diagonals of a quadrilateral bisect each other if and only if the quadrilateral is a parallelogram.

Unsurprisingly, we even have a symbol for 'if and only if', $\Leftrightarrow$:

The diagonals of a quadrilateral bisect each other $\Leftrightarrow$ the quadrilateral is a parallelogram.

Usually this symbol is only used in brief statements. For example: $x^2 = 4 \Leftrightarrow x = \pm 2$. Finally, sometimes the phrase 'if and only if' is shortened to 'iff.'

Exercises

Prove or disprove each of the following statements.

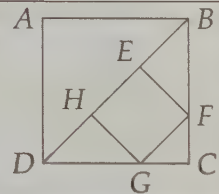
- 8.7.1 $EFGH$ is a rhombus if and only if its diagonals are perpendicular.
- 8.7.2 $\triangle XYZ$ is equilateral if and only if $XY = YZ$ and $\angle X = 60^\circ$.
- 8.7.3 A quadrilateral is a parallelogram if and only if it is a rhombus.
- 8.7.4 A rhombus is a square if and only if it is also a rectangle.
- 8.7.5 $PQRS$ is a parallelogram if and only if $PQ = RS$ and $SP = QR$.
- 8.7.6 Trapezoid $ABCD$ with $\overline{AB} \parallel \overline{CD}$ is isosceles if and only if $\angle ABD = \angle BAC$. **Hints:** 466
- 8.7.7 $WXYZ$ is a rectangle if and only if $WX = YZ$, $\angle WXY = \angle XYZ$, and $\overline{WX} \parallel \overline{YZ}$.
- 8.7.8 Let M be the midpoint of $\overline{AB}$. $\triangle ABC$ is isosceles with $AC = BC$ if and only if $\overline{CM}$ bisects $\angle ACB$. **Hints:** 4
- 8.7.9 $ABCD$ is a square if and only if its diagonals are perpendicular and congruent.

8.8★ Quadrilateral Problems

In this section we will tackle a few more challenging problems using the quadrilateral principles we have learned.

Problems

Problem 8.27: Diagonal $\overline{BD}$ of square $ABCD$ is drawn. Square $EFGH$ is then inscribed in $\triangle BDC$ with two vertices on $\overline{BD}$ as shown. If $AB = 6$, find the area of $EFGH$.



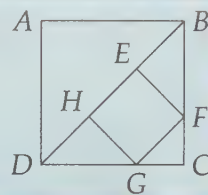
Problem 8.28: I have a rectangular painting that has a length 10 inches longer than its width. I frame it with a rectangular frame that is two inches wide all the way around. Given that the area of the frame is 152 square inches, what is the area of the painting?

Problem 8.29: Find the area of trapezoid $ABCD$ given that $\overline{AB} \parallel \overline{CD}$, $AB = 16$, $BC = 15$, $CD = 30$, and $AD = 13$.

Problem 8.30: Isosceles trapezoid $ABCD$ with $\overline{AB} \parallel \overline{CD}$ is inscribed in a circle with radius 10 such that the center of the circle is inside $ABCD$. Isosceles trapezoid $CDEF$, with $\overline{CD} \parallel \overline{EF}$, is inscribed in the same circle, but the center of the circle is not inside $CDEF$. Given that $AB = EF = 12$ and $CD = 16$, find the areas of $ABCD$ and $CDEF$.

Problem 8.31: Prove that the sum of the squares of the diagonals of a parallelogram equals the sum of the squares of the sides of the parallelogram.

Problem 8.27: Diagonal $\overline{BD}$ of square $ABCD$ is drawn. Square $EFGH$ is then inscribed in $\triangle BDC$ with two vertices on $\overline{BD}$ as shown. If $AB = 6$, find the area of $EFGH$.

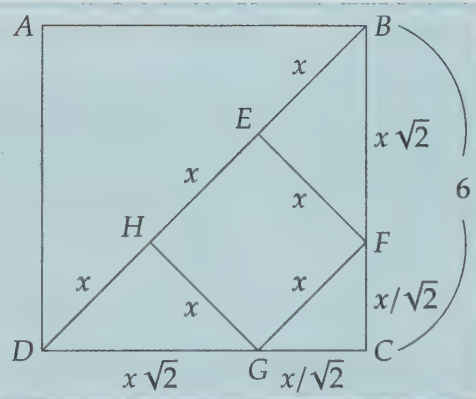


Solution for Problem 8.27: We need the side length of $EFGH$ to get its area, so we let the side length of $EFGH$ be x . Since $\triangle EBF$ is a 45-45-90 triangle, we have $BF = EF\sqrt{2} = x\sqrt{2}$. We also have $BE = EF = x$ (and from $\triangle DHG$, $DH = x$). Because $\overline{BD} \parallel \overline{FG}$, we have $\angle FGC = 45^\circ$, so $\triangle FCG$ is also a 45-45-90 triangle. Therefore, $FC = FG/\sqrt{2} = x/\sqrt{2} = x\sqrt{2}/2$. We know $BC = 6$, so we have $BF + FC = BC = 6$. Substitution gives

$$x\sqrt{2} + \frac{x\sqrt{2}}{2} = 6.$$

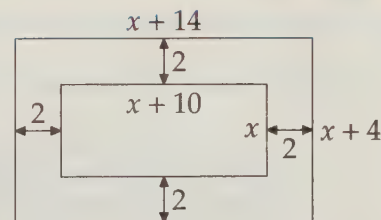
Therefore, $x\left(\frac{3\sqrt{2}}{2}\right) = 6$, so $x = 2\sqrt{2}$. So, the area of $EFGH$ is $EF^2 = x^2 = (2\sqrt{2})^2 = 8$. Alternatively, we could have noted that $BD = 3x = 6\sqrt{2}$, so $x = 2\sqrt{2}$. $\square$

Concept: Don't just stare at a diagram. Set sides you seek equal to variables and find other lengths in terms of those variables. Label the diagram when you find these new lengths. Your final diagram for the last problem might look like the one at right.



Problem 8.28: I have a rectangular painting that has a length 10 inches longer than its width. I frame it with a rectangular frame that is two inches wide all the way around. Given that the area of the frame is 152 square inches, what is the area of the painting?

Solution for Problem 8.28: We start by making a sketch of our painting with its frame. We let the width of the painting without frame be x , so the length is $x + 10$. Since the frame is two inches thick, we find that the painting with frame is $x + 14$ inches long and $x + 4$ inches wide. We can then find the area of the frame in terms of x by subtracting the small rectangle (the picture) from the large one (picture plus frame). The area



of the picture plus frame is $(x + 14)(x + 4)$ and the area of the picture alone is $x(x + 10)$, so we have:

$$(x + 14)(x + 4) - x(x + 10) = 152.$$

Expanding the left side gives $x^2 + 18x + 56 - x^2 - 10x = 152$, so $x = 12$.

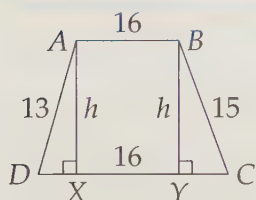
Now we can find the area of the painting without the frame: $x(x + 10) = 12(22) = 264$ square inches.

□

Concept: A picture is worth a thousand words – making a sketch can greatly help you understand a geometric problem.



Problem 8.29: Find the area of trapezoid $ABCD$ given that $\overline{AB} \parallel \overline{CD}$, $AB = 16$, $BC = 15$, $CD = 30$, and $AD = 13$.



Solution for Problem 8.29: We have the bases, so we just need the height. We draw altitudes from A and B , thus forming a rectangle and two right triangles. (We like right triangles!) Since $XY = AB = 16$ and $CD = 30$, we know that $DX + CY = 14$. At this point, there are a couple approaches we could take to find the height.

Wishful thinking. We have the hypotenuses of $\triangle AXD$ and $\triangle BYC$, and we know they each have a leg of length h . The 15 and 13 make us think of the Pythagorean triples 5-12-13 and 9-12-15. These each have a leg of length 12, so perhaps our height is 12. This would make $DX = 5$ and $CY = 9$, so $DX + CY = 14$, as desired. Our wishful thinking has been fruitful! The height must be 12, so we can now find the area.

Algebra. Perhaps we don't see the Pythagorean triples, or maybe they don't work out (the answers won't always be integers!); then, we have to do a little more work. We let $DX = x$, so $CY = 14 - x$. Now we can use the Pythagorean Theorem on $\triangle ADX$ and $\triangle BYC$ to build two equations in terms of h and x :

$$\begin{aligned} x^2 + h^2 &= 13^2 \\ (14 - x)^2 + h^2 &= 15^2 \end{aligned}$$

We can subtract the first from the second to eliminate h^2 :

$$(14 - x)^2 - x^2 = 15^2 - 13^2.$$

Factoring both sides as a difference of squares gives

$$(14 - x - x)(14 - x + x) = (15 - 13)(15 + 13),$$

so $(14 - 2x)(14) = 56$, which gives $x = 5$. Now we can find $h = 12$ with either of our original equations.

Finally, we find our area: $h(AB + CD)/2 = 276$. □

Concept: Rectangles and right triangles are easier to work with than trapezoids. When stuck on a trapezoid problem, consider dropping altitudes.



Problem 8.30: Isosceles trapezoid $ABCD$ with $\overline{AB} \parallel \overline{CD}$ is inscribed in a circle with radius 10 such that the center of the circle is inside $ABCD$. Isosceles trapezoid $CDEF$, with $\overline{CD} \parallel \overline{EF}$, is inscribed in the same circle, but the center of the circle is not inside $CDEF$. Given that $AB = EF = 12$ and $CD = 16$, find the areas of $ABCD$ and $CDEF$.

Solution for Problem 8.30: We'll tackle $ABCD$ first. In our diagram, we connect the center of the circle, O , to the vertices of the trapezoid, since we somehow want to use the fact that the trapezoid is inscribed in a circle. We have $OA = OB = OC = OD = 10$. We need a height, so we draw the height through O , since this will build right triangles in which we already know the hypotenuses. We let X and Y be the feet of the altitudes from O to $\overline{AB}$ and $\overline{CD}$, respectively.

Since $AO = BO$ and $OX = OX$, we have $\triangle AXO \cong \triangle BXO$ by HL Congruence. Therefore, X is the midpoint of $\overline{AB}$. Similarly, Y is the midpoint of $\overline{CD}$. Hence, $AX = 6$ and $DY = 8$. We can now use the Pythagorean Theorem on $\triangle XOA$ and $\triangle DOY$ to determine that $OX = 8$ and $OY = 6$. Thus, the height is $OX + OY = 14$, and our area is $(XY)(AB + CD)/2 = 196$.

Now we find the area of $CDEF$. We could break the problem into right triangles as before, or we could be a little more clever and apply what we learned about $\triangle AOB$ to $\triangle EOF$. Since $AO = EO$, $AB = EF$, and $BO = FO$, we have $\triangle AOB \cong \triangle EOF$. Therefore, the altitudes from O to $\overline{AB}$ and to $\overline{EF}$ are congruent. Letting Z be the foot of the altitude from O to $\overline{EF}$, we have $OZ = OX = 8$. Moreover, since $\overline{CD} \parallel \overline{EF}$ and $\overline{OZ} \perp \overline{EF}$, we have $\overline{OZ} \perp \overline{CD}$. Since Y is the foot of the altitude from O , $\overline{OZ}$ passes through Y . So, the height of $CDEF$ is $YZ = OZ - OY = 2$. Finally, the area of $CDEF$ is $(YZ)(CD + EF)/2 = 28$. $\square$

We'll finish with a challenging proof.

Problem 8.31: Prove that the sum of the squares of the diagonals of a parallelogram equals the sum of the squares of the sides of the parallelogram.

Solution for Problem 8.31: Let $ABCD$ be our parallelogram. We wish to prove

$$AC^2 + BD^2 = AB^2 + BC^2 + CD^2 + DA^2.$$

Since $AB = CD$ and $AD = BC$, we can simplify this to

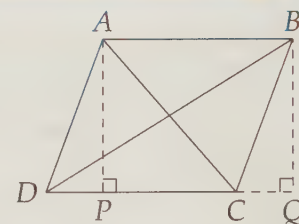
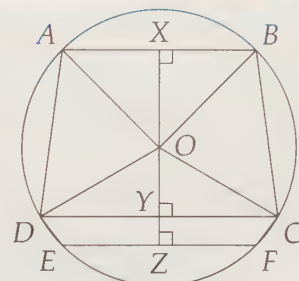
$$AC^2 + BD^2 = 2(AD^2 + CD^2).$$

Seeing the sums of squares of sides, we reach for the Pythagorean Theorem. We have no right triangles, so we'll have to build them. We draw altitudes from A and B , thus building right triangles with our diagonals as hypotenuses. From right triangles $\triangle APC$ and $\triangle BQD$, we have

$$\begin{aligned} AC^2 &= PC^2 + AP^2 \\ BD^2 &= DQ^2 + BQ^2 \end{aligned}$$

We can add these to get an expression for the sum of the squares of the diagonals:

$$AC^2 + BD^2 = PC^2 + AP^2 + DQ^2 + BQ^2.$$



That looks like what we want, but we're not quite there. We have to get AD and CD in the equation somehow. We start by noting that $DQ = CD + CQ$ and $PC = CD - DP$. We also note that $AP = BQ$ and $AD = BC$, so $\triangle ADP \cong \triangle BCQ$ by HL Congruence. Thus, $DP = CQ$. Now our equation becomes:

$$\begin{aligned} AC^2 + BD^2 &= PC^2 + AP^2 + DQ^2 + BQ^2 \\ &= (CD - DP)^2 + AP^2 + (CD + CQ)^2 + BQ^2 \\ &= CD^2 - 2(CD)(DP) + DP^2 + AP^2 + CD^2 + 2(CD)(CQ) + CQ^2 + BQ^2 \\ &= CD^2 - 2(CD)(DP) + DP^2 + AP^2 + CD^2 + 2(CD)(DP) + DP^2 + AP^2 \\ &= 2(CD^2) + 2(DP^2 + AP^2). \end{aligned}$$

Since $DP^2 + AP^2 = AD^2$ from right triangle $\triangle DAP$, we have the desired

$$AC^2 + BD^2 = 2(CD^2 + AD^2).$$

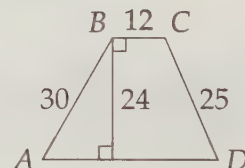
As we noted earlier, $CD = AB$ and $AD = BC$, so we've shown that the sum of the squares of the sides of a parallelogram equals the sum of the squares of its diagonals. $\square$

Exercises

8.8.1 In rectangle $ABCD$, H is the midpoint of $\overline{BC}$, E lies on $\overline{AD}$, and F lies on $\overline{AB}$. In rectangle $CEFG$, H lies on $\overline{FG}$ and $HG = 3$. Given $\angle DEC = 45^\circ$, what is the positive difference between the areas of these two rectangles?

8.8.2 Find the area of trapezoid $ABCD$ shown at right.

8.8.3 Quadrilateral $ABCD$ is a trapezoid with $\overline{AB} \parallel \overline{CD}$. We know $AB = 20$ and $CD = 12$. What is the ratio of the area of $\triangle ACB$ to the area of $ABCD$? (Source: MATHCOUNTS) **Hints:** 376

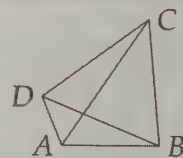


8.8.4 The diagonals of $EFGH$ are perpendicular. Prove that $EF^2 + GH^2 = FG^2 + EH^2$. (A quadrilateral with perpendicular diagonals is sometimes referred to as **orthodiagonal**.) **Hints:** 572

8.8.5★ In $\triangle ABC$, $AB = 6$, $BC = 7$, and $AC = 8$. Given that M is the midpoint of $\overline{AB}$, find CM . **Hints:** 240, 476, 117

8.9 Summary

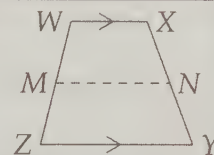
Definitions: A **quadrilateral**, such as $ABCD$ at the right, has four segments as sides, four vertices, and four angles. The segments connecting opposite vertices are called the **diagonals** of a quadrilateral.



Important: The interior angles of any quadrilateral add to 360° .



Definitions: A **trapezoid** is a quadrilateral with two parallel sides. The segment connecting the midpoints of the non-parallel sides is the **median** of the trapezoid, and the distance between the two parallel sides is the **height** of the trapezoid.



Important:

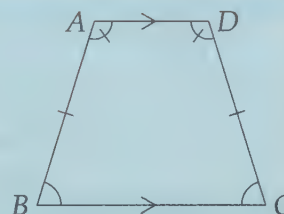


- The median of a trapezoid is parallel to the bases of a trapezoid, and equal in length to the average of the lengths of the bases.
- The area of a trapezoid equals the height of the trapezoid times the length of the median of the trapezoid.

Important: In an isosceles trapezoid:



- The base angles come in two pairs of equal angles as shown at right.
- The legs are equal.
- The diagonals are equal.



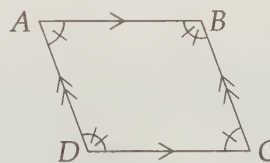
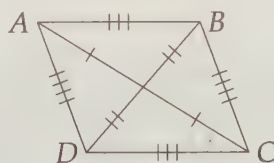
If any one of these items is true for a trapezoid with exactly one pair of parallel sides, then all the others must be true for that trapezoid.

Definitions: A **parallelogram** is a quadrilateral in which both pairs of opposite sides are parallel.

Important:



- The area of a parallelogram is the product of a side length (the base) and the distance between that side and the opposite side of the parallelogram. This distance between opposite sides is called a **height** of the parallelogram.
- In parallelogram $ABCD$, the opposite sides are equal, the opposite angles are equal, and the diagonals bisect each other.



Conversely, $ABCD$ is a parallelogram if *any one* of the following are true:

- $AB = CD$ and $AD = BC$.
- $\angle A = \angle C$ and $\angle B = \angle D$.
- Diagonals $\overline{AC}$ and $\overline{BD}$ bisect each other.

Therefore, proving one of these means the other two are true.

Definition: A quadrilateral is a **rhombus** if all of its sides are equal.

Important:



- Every rhombus is a parallelogram. Therefore, everything that is true about parallelograms is true about rhombi.
- The diagonals of a rhombus are perpendicular. The area of a rhombus is half the product of its diagonals (and also equals its base times its height).

Definition: A quadrilateral in which all angles are equal is a **rectangle**.

Important:



- All rectangles are parallelograms, so all that is true of parallelograms is true of rectangles.
- Let two consecutive sides of a rectangle have lengths ℓ and w . The area of the rectangle is ℓw , and the diagonals of the rectangle both have length $\sqrt{\ell^2 + w^2}$.

Definition: A quadrilateral in which all sides are equal and all angles are equal is a **square**.

Important:



- Each square is a parallelogram, a rectangle, and a rhombus so all that is true of a parallelogram, a rectangle, or a rhombus is true of a square.
- If the side length of a square is s and its diagonal is d , then $d = s\sqrt{2}$ and the area of the square is s^2 , or $d^2/2$.

In proving various facts about quadrilaterals, we encountered the phrase ‘if and only if.’ Proving ‘if and only if’ statements requires proving two different statements.

Problem Solving Strategies

Concepts:



- Triangles, triangles, triangles. Although this chapter was about quadrilaterals, looking back you’ll see that a great many of our solutions revolved around breaking the problems into triangles on which we can use all our triangle tools.
- Rectangles and right triangles are easier to work with than trapezoids. When stuck on a trapezoid problem, consider dropping altitudes to build rectangles and right triangles.

Continued on the next page. . .

Concepts: . . . continued from the previous page



- Breaking a desired angle, length, or area into parts often makes it easier to find.
- When stuck on a problem, think ‘What information haven’t I used yet?’ Then, try to find some way to use that information.
- Don’t just stare at a diagram. Set sides you seek equal to variables and find other lengths in terms of those variables. Label the diagram when you find these new lengths.
- A picture is worth a thousand words – making a sketch can greatly help you understand a visual problem.

Things To Watch Out For!

WARNING!!



- Although every rhombus is a parallelogram, every parallelogram is **not** a rhombus. Therefore, if we prove a property of rhombi, this property is not necessarily true for parallelograms. (The same is true of rectangles – rectangles are parallelograms, but not all parallelograms are rectangles, etc.)
- Be very careful that you do not assume what you are trying to prove as part of a proof!

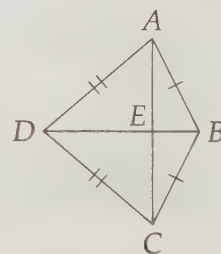
REVIEW PROBLEMS

8.32 Fill in each of the boxes with the answer “Yes”, “Sometimes”, or “No”.

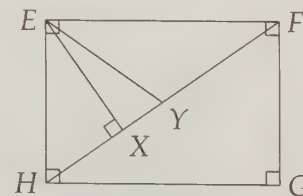
Fact	Parallelogram	Rhombus	Rectangle	Square
Opposite sides have equal length				
All sides have equal length				
Opposite sides are parallel				
Opposite angles are equal				
All angles are equal				
Diagonals bisect each other				
Diagonals have equal length				
Area can be determined from sides				

- 8.33 In quadrilateral $ABCD$, $\angle A = \angle B = 128^\circ$, and $\angle D$ is 10° less than 5 times $\angle C$. Find $\angle C$ and $\angle D$.
- 8.34 In isosceles trapezoid $ABCD$, $\angle A = x$ and $\angle B = 2x - 45^\circ$. Find $\angle C$ and $\angle D$ in the following cases:
- $\overline{AB} \parallel \overline{CD}$.
 - $\overline{AD} \parallel \overline{BC}$.
- 8.35 In which of the following quadrilaterals does a diagonal always divide the quadrilateral into two regions of equal area: Rhombus, Square, Rectangle, Trapezoid, Parallelogram?
- 8.36 If the numbers denoting the perimeter and area of a square are equal, what is the length of its diagonal?
- 8.37 If A, B, C, D are midpoints of the sides of square $EFGH$, what is the ratio of the area of triangle ABC to the area of square $EFGH$?
- 8.38 The bases of a trapezoid are $2x$ and $4x$. The height is $2x$. If the area is 48, what is x ?
- 8.39 One of the angles of a rhombus is 120° . If the shorter diagonal is 2, what is the area?
- 8.40 What is the number of centimeters in the length of a longer side of a rectangle that has a perimeter of 64 centimeters and an area of 192 square centimeters?
- 8.41 Quadrilateral $WXYZ$ is a parallelogram. Given that $WX = 2x - 3$, $XY = x + 7$, and $YZ = 3x - 8$, find the perimeter of $WXYZ$.
- 8.42 $WXYZ$ is a rhombus with $\angle X = 90^\circ$. Prove that $WXYZ$ is a square.

8.43 One less well-known type of quadrilateral is the **kite**. The four sides of a kite can be split into two pairs of consecutive equal sides. For example, the figure at right shows kite $ABCD$ with $AB = BC$ and $CD = DA$. The diagonals of $ABCD$ meet at E as shown. Solve the following problems about kite $ABCD$.



- Prove that $\angle ABD = \angle CBD$.
 - Prove that $\overline{AC} \perp \overline{BD}$. **Hints:** 140, 249
- 8.44 Which of the following quadrilaterals is (are) a special case of a kite: parallelogram, trapezoid, rhombus, rectangle, square? (More than one answer may be correct.)
- 8.45 Each side of an equilateral triangle is 8 inches long. An altitude of this triangle is used as the side of a square. What is the number of square inches in the area of the square? (Source: MATHCOUNTS)
- 8.46 Shown at right is rectangle $EFGH$. Given that $\angle GHF = 31^\circ$ and $EY = HY$, solve the following problems:
- Find $\angle XEH$.
 - Find $\angle YEF$.
 - ★ Prove that $\overrightarrow{EY}$ passes through G . **Hints:** 366, 214
- 8.47 $EFGH$ is a rhombus. Given that $EF = EG = 6$, find the area of the rhombus. **Hints:** 404



8.48 Prove or disprove this statement: $ABCD$ is a square if and only if its sides are equal in length and $AC = BD$.

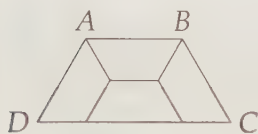
8.49 The angles of quadrilateral $WXYZ$ are such that $\angle W > \angle X > \angle Y > \angle Z$. The angles are in an arithmetic progression, meaning that $\angle W - \angle X = \angle X - \angle Y = \angle Y - \angle Z$.

(a) If $\angle W$ is four times the measure of $\angle Z$, what is $\angle W$?

(b) If $\angle W$ is two times $\angle Z$, what is $\angle W$?

8.50 In quadrilateral $ABCD$, let E , F , G , and H be midpoints of $\overline{AB}$, $\overline{BC}$, $\overline{CD}$, and $\overline{DA}$, respectively. Prove that $EFGH$ is a parallelogram.

8.51 A square and an equilateral triangle have the same perimeter. What is the ratio of the area of the square to the area of the equilateral triangle?



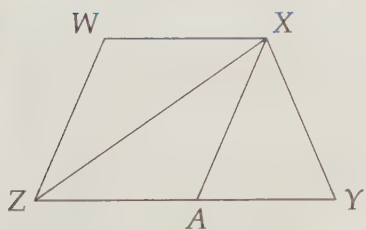
8.52 Trapezoid $ABCD$ is divided into four congruent trapezoids as shown. Given $AB = 4$ and $CD = 8$, find the sum of the lengths of all line segments in the figure. (Source: ARML)

8.53 What is the ratio of the area of the shaded square to the area of the large square in the figure at right? (The figure is drawn to scale.) (Source: AMC 8)



8.54 A street has parallel curbs 40 feet apart. A crosswalk bounded by two parallel stripes crosses the street at an angle. The length of the curb between the stripes is 15 feet and each stripe is 50 feet long. Find the distance between the stripes. (Source: AMC 10)

Challenge Problems



8.55 In the diagram at left, $AX = XY = 6$, $\angle AXY = 70^\circ$, $\angle W = 125^\circ$, $\angle XZA = 21^\circ$, and $\angle XZW = 34^\circ$.

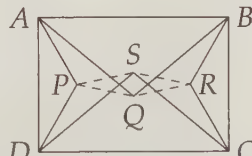
(a) Find $\angle ZXA$.

(b) Find WZ .

(c) Prove that $WY = XZ$. **Hints:** 328, 558

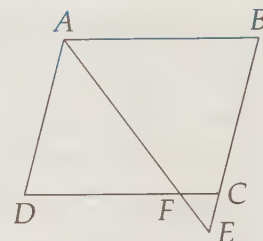
8.56 In rectangle $ABCD$, $AB = 16$ and $AD = 5$. F is on $\overline{AB}$ and G on $\overline{CD}$ such that $\overline{FG}$, $\overline{BD}$, and $\overline{AC}$ are concurrent at point O . Find $[FOB] + [GOC]$. (Source: MATHCOUNTS) **Hints:** 190

8.57 Each angle of rectangle $ABCD$ is trisected by a pair of segments. The angle trisectors meet at P , Q , R , and S , as shown in the diagram at right. Prove that $PQRS$ is a rhombus. **Hints:** 243



8.58 In rectangle $ABCD$, M is the midpoint of $\overline{BC}$. Points P and Q lie on $\overline{AB}$ and $\overline{DC}$, respectively, such that $PB = \frac{4}{3} \cdot BC$ and $\angle PMQ$ is a right angle. What is the ratio $PM : MQ$? (Source: MATHCOUNTS) **Hints:** 306

8.59 In the diagram, $ABCD$ is a parallelogram and E is on the extension of $\overline{BC}$ past C . $\overline{AE}$ and $\overline{CD}$ meet at F . Given $[ADF] = 64$ and $[CEF] = 4$, determine the following:



- (a) CF/DF .
- (b) $[BFC]$. **Hints:** 373
- (c) $[ABCD]$. **Hints:** 213

(Source: ARML)

8.60 See Problem 8.43 for the definition of a kite.

- (a) Find the area of a kite that has diagonals of length 10 and 12. **Hints:** 118, 299
- (b)★ Find the area of kite $ABCD$ if $AC = 6\sqrt{2}$, $AB = 4\sqrt{3}$, and $BC = 2\sqrt{3}$. **Hints:** 193, 18, 527

8.61 Consider a square of side length 1. We draw four lines that each connect a midpoint of a side with a corner not on that side, such that each midpoint and each corner is touched by only one of these lines as shown at left below. Find the area of the shaded region. (Source: HMMT) **Hints:** 303

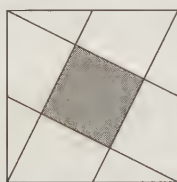


Figure 8.9: Diagram for Problem 8.61

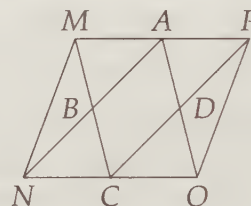


Figure 8.10: Diagram for Problem 8.62

8.62 Given $\overline{MP} \parallel \overline{NO}$, $\overline{PO} \parallel \overline{MN}$, $MA = AP$, and $NC = CO$ as in the diagram at right above, prove that $[ABCD] = [MNOP]/4$. **Hints:** 386

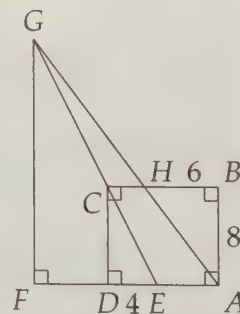
8.63 The diagonals of $WXYZ$ meet at A . Prove that the quadrilateral formed by connecting the incenters of $\triangle WXA$, $\triangle XYA$, $\triangle YZA$, and $\triangle ZWA$ has perpendicular diagonals. **Hints:** 368, 459

8.64 $EFGH$ is a rectangle with $EF = 12$ and area 192. $EGJI$ is a parallelogram such that H is on $\overline{JI}$. What are the possible values for the area of $EGJI$? **Hints:** 486

8.65 In rectangle $ABCD$, we have $AB = 8$, $BC = 9$, H is on $\overline{BC}$ with $BH = 6$, E is on $\overline{AD}$ with $DE = 4$, line $\overleftrightarrow{EC}$ intersects line $\overleftrightarrow{AH}$ at G , and F is on line $\overleftrightarrow{AD}$ with $\overline{GF} \perp \overline{AF}$. Find the length of $\overline{GF}$. (Source: AMC 10) **Hints:** 521, 379

8.66 In trapezoid $ABCD$, $\overline{AD} \parallel \overline{BC}$. $\angle A = \angle D = 45^\circ$, while $\angle B = \angle C = 135^\circ$. If $AB = 6$ and the area of $ABCD$ is 30, find BC . (Source: HMMT) **Hints:** 552

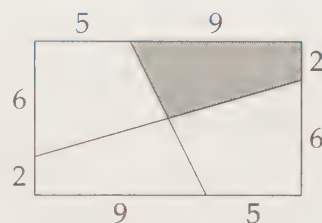
8.67 A circle is drawn through the vertices of square $ABCD$, and point X is on minor arc $\widehat{AB}$. Given that $[XAB] = 1$ and $[XCD] = 993$, find $[XAD] + [XBC]$. (Source: Mandelbrot) **Hints:** 412



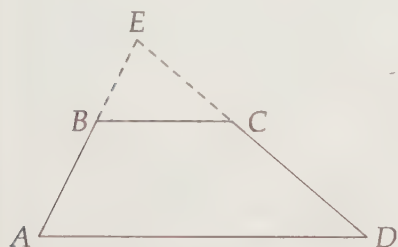
Extra! A pupil from whom nothing is ever demanded that he cannot do, never does all he can.

—John Stuart Mill

8.68★ The lengths indicated on the rectangle shown are in centimeters. What is the number of square centimeters in the area of the shaded region? (Source: MATHCOUNTS) **Hints:** 516

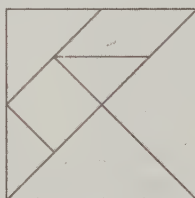


8.69★ P is inside rectangle $ABCD$. $PA = 2$, $PB = 3$, and $PC = 10$. Find PD . (Source: HMMT) **Hints:** 247, 381

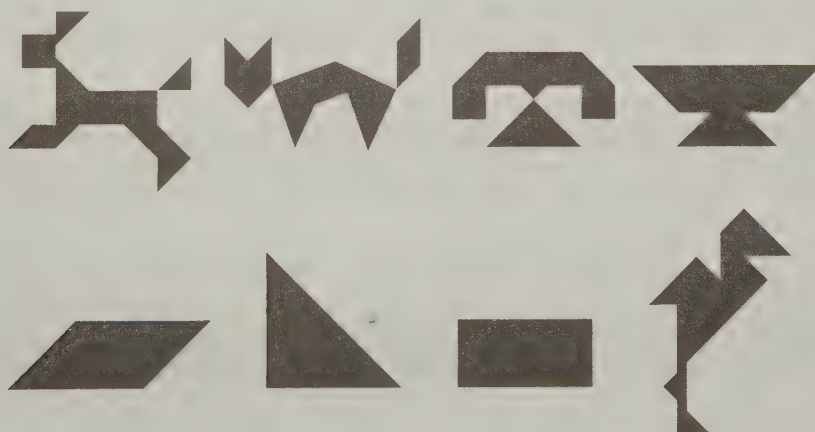


8.70★ In trapezoid $ABCD$ shown, $\overline{AD} \parallel \overline{BC}$, $AB = 6$, $BC = 7$, $CD = 8$, and $AD = 17$. The sides $\overline{AB}$ and $\overline{CD}$ are extended to meet at E . Prove that $\angle E = 90^\circ$. (Source: HMMT) **Hints:** 511

Extra! One of the oldest known puzzles is the Chinese *ch'i ch'iao t'u*, which means 'ingenious seven-piece plan.' In English, these are known as **tangrams**.



Tangrams involve rearranging the seven pieces that together make the square shown above into other given shapes. Some shapes that can be formed by these pieces are shown below. See if you can figure out how!



Perhaps you'll notice a couple things tangrams have in common with math problems. First, with experience, they get a lot easier. Second, often the simplest looking shapes end up being the hardest ones to solve! Source: *Martin Gardner's Mathematical Puzzles & Diversions*